BT 6003 - Principles of Food processing – Semester V

UNIT - I

Constituents of food

Dietary carbohydrates and their sources

- Carbohydrates occur in food as sugars and starch, which are the major sources of energy in the diet and as cellulose, the main component of dietary fibre.
- The carbohydrates composition in foods influences flavour, texture and colour and thereby the acceptability of foods.

Common carbohydrates and their sources

Carbohydrates	Sources
Monsacccharides Glucose and fructose	Honey, fruits
Oligosaccharies 1) raffinose, stachyose 2) Fructosyl suchrose and sucrose 3) Lactose 4) Maltose, isomaltose	Cerials, onion, malt, tubers Sugarcane, fruits, vegetables Milk and other diary products Starch syrup, honeycell walls, cereals , malt
Polysaccharides 1) Starch and dextrins 2) Cellulose 3) Glycogen 4) Hemicellulose	Cereals, roots, tubers, legumes Plant cell wall and fibres Liver and animal tissues Plant

- Fruits such as apples, bananas and pears are harvested before they completely ripen and the starch is covert to sugars as ripening proceeds during storage.
- Green peas, green beans and sweet corn are harvested before maturity to obtain sufficent textures and sweetness.
- As maturation progresses, the sugars are converted into polysaccharides, water is lot and tough textures develop.

Carbohydrate derivatives

- Uronic acids are important constituents of polysaccharides eg glucoronic acid and galacturonic acid.
- Reduction of sugar leads to sorbitol
- Sorbitol is found in nature in fruits such as pears, apples and plums.
- Inositol is found in phytin in plants.

- These sugar alcohols are not absorbed by the human intestine.
- Hence these sugar alcohols are used as sweeteners

Nutritional aspects

- Carbohydrate are the cheapest source of energy in the human diet providing about 4 k Cal/g.
- Apart from their nutritional values, carbohydrates contribute to the appetizing and pleasurable aspects of food such as sweetness and flavous due to caramelization.
- Glucose, sucrose, maltose, lactose, glactose, fructose and ribose are digestible readily in human.
- Melibiose, mannose, sorbose, arabinose and xylose either remain undigested or poorly abosrbed in human metabolism
- In nutritional point of view, starch and glycogen is the major polysaccharide that are readily digested in the intestine and thus serves as a sources of carbohydrates.
- The energy requirement of body is met by the oxidation of glucose to 2 mol of pyruvic acid in glycolysis releases about 8 k. Cal of energy followed by the oxidation of pyruvic acid to carbondioxide and water producing about 3 k.cal

Flavour and colour production by carbohydrates

- Flavour and colour reactions of carbohydrate constituents of foods may be classified into
  - Enzymic browing reactions
  - Caramelization
  - Millard reaction and
  - Strecker degradation

The last three reactions are non- enzmatic browing reaction whaicha re responsible for flavour and colour development in foods

Enzymatic browning reactions

- The browning reactions that occur in foods may be classified into two types
  - Oxidative browning
  - Non oxidative browning

Oxidative browning, oxidases are active in frist step in conversion of phenol enediol or conjugated diene to reactive carbonyl compounds which undergo further non enzymic oxidative browning reactions

- Non oxidative browning also involves enzymes in the first step in which glycosyl hydrolases initiate the so called non oxidative browning in natural products by the release of reducing sugars.
- This is an important step in colour and flavour development in dates, honey maple syrup, c hoclate and vanilla processing

Caramelization

- Decomposition reactions may be induced by melting dry sugars and to a limited extent by heating their syrups at temperuture about 100 C.
- Heat hastens the inversions of sucrose to yield reducing sugars which condense to higher oligosaccharides and dextrins.
- The brown pigments that characterize caramel and other food are due to HMF and its precursors.
- The pigment molecules have very high molecular weight are are not absorbed in the intestine

Maillard reaction

- In the presences of amino compounds the browning of sugars occurs more rapidly, particularly in neutral or alkaline conditions in a sequence of reactons called Maillard reaction.
- Sugar amine reactions result in amorphous brown to black polymers called melanoidins.
- Browning and aroma formation during cooking, frying, backing or roasting are essentially caused by the Maillard reaction
- Maillard reaction results in a loss of amino acids. Lysine is the most reactive of amino acids followed by methionine, arginine, tryptophan and histidine.
- Since lysine , histidine and tryptophan are essential amino acid.
- It is important to retard the Millard reaction in foods.

- Inhibition of maillard reaction is accomplished by keeping the pH below the isoelectric points of amino acids, use of low temperatures and by diluting with water

Strecker Degradation

- An important class of volatile flavours arises by an interaction of α- dicarbonyl compounds (end or intermediate products of catabolic reaction) with α amino acids at elevated temperatures called the Strecker degradation reaction which involves transmination and decarboxylation reaction.
- The degradation reaction yields aldehydes, aminoketones and carbon dioxide.

Texture characteristics of carbohydrates

- Monosaccharide, oligosaccharides and particularly, polysaccharides immensely influence textural characteristics of foods.
- Sugars in their crystalline state contribute to the appearance and texture of foods products, particularly, the confectionery, biscuits and cakes.
- The granulated sugar and powdered form for icing available are commercially available.

Sugars

- Sugars form super saturated syrups when their aqueous solutions are concentrated.
- The syrups of reducing sugars are very resistant to crystallization.
- Very high dissolved sugar concetration are essential feature of jam and other preserves.
- In preserves, water activity is reduced by hydrogen bonding with sugar molecules thereby preventing the growth of micoorganisms.

Starch

- The food texture in relation to molecular structure of starch is quite interesting
- Starch occurs in abundance in the form of irregular round shaped aggregates or granules, ranging in size from 2-100 μm.
- The shapre and size of the graanules are peculiarity of the species of the plant and can be used to identify the orgin of starch of flour.

- Starch consists of two types amylose and amylopectin
- Most starches have 20-25% amylose.
- Pea starcch is 60% amylose.
- Waxy varieties of maize and other cereals have very little or no amylose.
- Raw uncooked starchy foods are not easy to digest because starch granules are contained within the cell walls of the plant which cannot be penetrated easily by the digestive juices.
- Cooking softens the cell walls and alows water to enter starch granules causing them to disintegrate and gelatinize

Pectin

- Pectin (poly galactouronic acid) occurs in the middle lamella of pant tissues and in the primary cell wall.
- It is the structually polysaccharide of soft tissues in soft fruits and fleshy roots.
- pectin forms gels that give the texture of jam and fruit preserves.
- It is produced from peels of citrus from apple pomace by extraction at pH 1.5-3.0 at 60-100 C

Agar

- Agar is obtained from sea weed by hot water extraction.
- The main components of the polysaccharide are beta D glactopyranose and 3,6 anhydro -α-L-galactopyranose
- Agar is insoluble in cold water but may be solubilized by precepitating it from a warm aqeuous dispersion by the addition of ethanol
- Agar is good gelling agent and the gel setting and stability depend on the concentration and molecular weight.
- Agar finds use as a stabilizer and emulsifier in frozen desserts of fruit juice, ice creams, yoghurt, cheese, candy and bakery products.
- It retards the staling of bread and also provides the desired texture in canned meat and poultry products.

Alginates

- Alginates occur as skeletal components of cell walls of brown algae.

- Alginates are extracted with alkali and precipated by acids or calcium salts.
- Alginate is used in food industry as thicking, stabilizing and gel forming agent. It stabilizes the consistency of filling the baked products salads dressing and milk chocolates.

PROTEINS:

They are macro molecules
Mol wt ranges from 10,000 to several hundred thousand
Required for nutrition and textural quality of food
Imbalance leads to malnutrition.

Classification of Proteins

- On basis of solubility :
  - Albumins – water soluble
  - Globulin – insoluble in water but soluble in salt solution
  - Glutelins – soluble in acids or bases
  - Prolamines – soluble in 70% alcohol water mixture.
- On basis of functional role in biological system
  - Carrier proteins : haemoglobin & permeases
  - Enzymes : biocatalyst
  - Immunoglobins : form antibodies to fight against invading microorganisms
  - Structural proteins : collagen
  - Proteins with nutritional functions : casein

Role in biological systems

Sources of dietary amino acids, growth and maintenance of living system.
Action of hydrolytic enzymes helps in break down of complex proteins into amino acids.
On absorption, they become a part of the amino acid pool
Non essential and essential amino acids are present.

Dietary sources and functions in food

Eggs , milk, meat
Legumes such as soy and peas

Functions :

1. Swelling and gelation : depends on solubility of protein which depends on conformation and intermolecular interactions at a given pH and temp. gelation contributes to the viscosity of the solution.

2. Water holding capacity of muscle protein influences the taste, tenderness and colour of the meat

3. Coagulation of milk gives casein, responsible for texturing.

4. Formation of stable foams in whipped creams and ice creams

5. Emulsification and structure formations are provided by egg proteins

6. Antibacterial activity such as nicin to prevent spore forming bacteria helpful in canning and cheese making

Modification of functional properties of proteins

Modification of proteins by chemical and enzymatic methods is gaining importance as modified proteins have better feature than native proteins.

Protein modification aims at

1. Blocking undesirable reactions such as maillard reaction.
2. Improving physical properties such as foam stability, solubility and texture
3. Increasing the nutritional value

Chemical modifications

1. Acylation of free amino groups
2. Reductive alkylation of amino groups with formaldehyde to give dimethylamino moiety.
3. Hydrolysis of amides to give carboxylic acids
4. Oxidation of thiol to disulphide bridges
5. Esterification of carboxylic and alcohol containing.

Enzymatic modification

1. Plastein reaction enables formation of larger polypeptides using papain
2. It is also used to improve the solubility of proteins and to remove undesirable amino acids.
3. Texturing in processed food such as meat
4. Soya, casein, wheat gluten, oilseed meals, sunflower, rapeseed, zein (corn protein), blood plasma etc, are suitable for texturization
5. Texturization is carried out by spin process or extrusion process

Spin process

Soya, protein isolate is extracted from soya flour in alkali to form viscous solution forced through tiny holes of spinneret to give fine threads of spun material in 10% NaCl and an acid bath.bundles pressed between rollers to remove coagulating solvents and neutralized to pH 5-6 in a bath of sodiun bicarbonate and NaCl. Aggreagtes formed are treated with additives such as egg protein, starch,lipids and are finally dried and sliced and marketed as spun vegetable protein (SVP)

Extrusion process

soya flour is moistened slightly with solution of NaCl, aroma compounds and buffers and made to a dough heated to 120 – 180°C undera pressure of 30 – 40 bars and extruded through a nozzle at atm pressure mass cut into pieces and dried and marketed as Textured vegetable protein (TVP)

Physical and chemical properties

Physical

ionic property : influences solubility.

Isoelectric point : very close, protein insoluble in water due to hydrophobic interactions and agglomeration.

Increase in salt conc upto 0.2M increases protein solubility

Chemical

Denaturation, maillard reaction, hydrolysis

Hydrolysis is done by action of acids , alkali and protoelytic enzymes

Milk

It is an aq solution of proteins, lactose, minerals, vitamins, emulsified fat globules.

Skimmed milk is the one from which fats have been removed & whey is obtained by removing casein from milk

Caseins are a group of associated phospho proteins which are heat stable

α & β casein form hydrophobic core of sub micelles where as κ casein stabilizes the casein

Milk includes alkaline phosphatase, lipase , protease, xanthin oxidase

Milk products include cheese , yoghurt, whole and skimmed milk powders.

Caseinates find importance because of their solubility at neutral pH

Milk whey powder contains upto 70% lactose

Denatured whole whey protein obtained by heating to 900C is called lactalbumin.

Meat

Red meat is the meat obtained from cattle, sheep, pig.

Fundamental unit of meat is myofibrillar sarcomere.

Rigor mortis – condition in which muscle is not able to relax after contraction or vice verse as in case of death.

Toughness can be avoided by holding the carcases between 35 – 400C

tenderness of meat determined by cooking.

Texture of meat is due to collagen. Collagen is usually insoluble and tough but due to moisture present, it is converted to gelatin.

Beef, lamb contains saturated fats whereas chicken contains polyunsaturated fats

Red colour of meat is due to collagen present

Fish

Flesh of fish is composed of fibres called myomeres.

Cooked fish os easily digested cause the elastin is absent.

Flesh of white fish such as cod, haddock contains very little fat

Fat fish such as herring ,trout, salmon contains 10- 25 % fat

It’s a rich source of proteins

Eggs

Egg consists of 3 parts: shell, egg white, yolk.
Main protein present is ovalalbumin.
Egg white protein is called albumin.
Nutritional value of egg is due to presence of Fe, P, fats, vit A, Ca.
Egg white is used in cake making.
Used as thickening agents and binding agents
Good emulsifiers

Cereal Proteins

Cereals include wheat, barley, maize, rice, etc.
These are used in bread and related items making.
Flours obtained from spring sown wheat are good for bread making.
Properties of wheat flour can be improved by prolonged storage.
During storage, auto oxidation of poly unsaturated acids results in formation of hydrogen peroxides which are good oxidising agents.

Seed proteins

Most plant seeds contains 15% of protein content as storage proteins.
Soya, peanut, dried peas and beans, sunflower, cotton seeds are useful.
Gelatin property of soya is useful in manufacture of canned meat, soup, and dairy products.
Cotton seed oil contains globulins in which 2S fraction is composed of low mol wt. 7S and 12S have high mol wt.

The role of organoleptic inspection in food qualification

Food quality components:

- Food hygiene properties
- Physical properties
- Chemical properties
- Packing and labelling
- *Organoleptic properties*
  - Exterior characteristics: surface, colour, packing, shape, form
  - Odour
  - Taste
  - Temperature
  - Substance and structure

Organs

- eyes, nose, tongue and mouth, fingers, ears(!)
- Using the human organs of senses as measurement tools raises some problems:
  - the human organs of senses are liable to sensorial errors
  - the human organs of senses are fatigable
  - there is a big biological variability
    - some bitter substances could be percepted only in higher concentration by some people
    - some people are sensitive to the chumestrol-content of the carrot

Sense of Taste

- ~6 million receptors on the tongue (dynamic balance),
  with aging death>production of receptors → ~1,5-2 million
- 4 basic tastes (sweet, salty, sour, bitter), but there are some more tastes:
  - taste generating pain sensation (capsaicine)
  - taste influencing the sensation of substance characteristics (alkaline materials or contractive materials like alum)
  - metallic taste ($FeSO_4 \times 7H_2O$)
  - the "umami" taste (complex taste, the taste of Na-glutamate; it's salty, sweet, bitter and metallic at the same time)
- Haller's taste catalogue contains 12 basic tastes (for example there is taste generating changes in heat perception – menthol)

Sweet taste

- natural (sugars; mollenin – 2000 times sweeter than sucrose; miraculum – changes the taste effect: makes sweet from sour)
- artificial (sugar-alcohols, ketones, esters, etc.) – they were developed during the research of diabetes
- Intensity of the sweet taste increases with water-solubility.
- Making symmetric structure from an asymmetric, the sweet taste changes into bitter

Sour, salty taste

- Sour taste: acids

- *Depends from:* acid concentration, pH, dissociation rate, other food components, the buffer effect of saliva
- *Intensity:* acetic acid (the least intensive)→lactic acid→malic acid→citric acid → tartaric acid (most intensive)

- Around the perception threshold (0,02g/100cm3) the typical salty substances (like NaCl) seem to be sweet, they appear to be salty only around recognizing threshold (0,08g/100cm3).

Salty taste: salt like materials

substance	taste	substance	taste
$MgCl_2$	salty-bitter	K_2SO_4	sour-bitter
NH_4Cl	salty-sour	$BeSO_4$	sweet-sour
$NaHCO_3$	salty-sweet	Pb-acetate	sweet (!)
$MgSO_4$	bitter-sweet	Salts of heavy metals	metallic

Bitter taste

- Substances containing nitrogen, inorganic salts, tanning materials, alkaloids (caffeine, quinine, morphine, nicotine), etc
- "Taste blindness": some people (and monkeys) don't percept the bitter taste.

Factors influencing personal values of taste sensing threshold

- pH of the sample
- temperature of the sample (no refrigeration temperature!)
- clearness of the aroma substance
- the water medium
- size of the tongue surface
- number of receptors on unit surface
- Physical and psychical condition of the judge (judge has to know his program 2 days before the inspection – no coffee, smoking, big breakfast at the morning,etc.)
- Circumstances of the inspection (noise, light, temperature, colour of the wall, etc.)
- part of the day (a.m. is the best – between 10-11 hours)
- experience of the judge (Can be learned and improved by practicing)
- method of the inspection (drop-method, spoon-method, etc.)

- other factors

Sense of Smell

- *Aroma:* odour (in mouth) + odour (in nose) + taste
- *Scent:* pleasant smell
- Smell materials: common characteristics are: the –OH, –CO, –COOH or –NH2 group
- Factors influencing the sense of smell: temperature (optimal: 25-30°C), part of the day (a.m.), age (20-40 years), sex of the judge (female), smoking (no), experience (high)
- Anomalies of the sense of smell:
  - *hypozmia*: decreased sense of smell
  - *hyperozmia:* hyper sensitivity (women)
  - *anozmia:* no sense of smell (above 80 years of age)
  - *merozmia:* no perception of specific smells
  - *autozmia:* sensation of smell without smelling substance

Textural characteristics

- Rheological classification of food:
  - liquid
  - cell tissue structure (vegetables, fruits)
  - gel structure (cheese, jelly)
  - fibre structure (meat and meat products)
  - plastic (butter, margarine)
  - hard, fragile (cookies, chocolate)

complex (bread)

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UNIT II

FOOD ADDITIVES

INTENTIONAL FOOD ADDITIVES

Those substances purposely added to perform specific functions includes

Preservatives, anti-oxidants, sequestrants, surface active agents, stabilizers and thickners, bleaching and maturing agents, starch modifiers, buffers, acids and alkalies, food colours, dietary sweenteners, flavouring agents etc.

Acids, Bases and their sals: Organic and inorganic acids of plant origin are deliberately added in food processing. Eg Citric acid (lemon , tomato), malic acid (apple and tomato), oxalic acid (lettuce), tartaric acid (grape and pineapple) Milk is considered as a complex buffer containing proteins, phosphate, citrate and calcium.

Monocarboxylic acids such as lactic acid, acetic acid induces coagulation of milk in cheese making. Lactic acid also used in prevention of discoloration of fruits and vegetables. Malic acid used in jellies and canned beverages. Tartaric acid used in fruit juice and sour candies.

Dicarboxylic acids such as Adipic acid used in baking powder and fruit juice powder. Succinic acid used in dehydrated food products. Fumaric acid increases the shelf life of jelly powders and pudding. Bezoic acid, sorbic acid used as antimicrobial agents. Phosporic acid used as acidulant in carbonated beverages such as cola and root beer.

Bases such as sodium bicarbonate used in peanut brittle candy to enhance sugar amino browing and for porous texture and for bitter chocolate flavor. Phosphates increase water binding capacity of raw and cooked meat and sausages.

Antioxidants: Prevent oxidation of fats in foods such as potato chips, cereals, salted nuts. Butylated hydroxyanisole (BHA), Butylated hydroxytolunene (BHT), propyl gallate (PG), ascorbic acid, tocopherol, (vitamine E) also been used.

Sequestrants: Traces of metals such as iron and copper catalyses undesirable oxidation reactions in foods.Sequestrants (metal scavengers) combine with metals remove from solution. EDTA, Polyphosphates are added in canned sea foods to prevent metal induced oxidation.

Stabilizers, thickneners and gel builders: Hydrocolloid material (Arabic gum, guar gum, carboxy methyl cellulose carageenan, agar, starch dextrins, gelatin, pectin) used in increase viscosity of gravies and fillings. They stabilize foam texture, emulsion in jellies, icings and chocolate milk drinks.

Surface acitive agenst: Emulsifiers (lecithin, mono, di glycerides) used in stabilize oil in water or water in oil mixtures.

Maturing agents and Starch modifiers: Freshly milled flour has yellowish tint due to carotenoid pigments and gives sticky dough properties during baking. Benzoyl peroxide, bromated and iodate are used to contion bread dough for baking performances. Calcium stearyl lactylate improve the dough mixing and loaf volume of bread.

Leavening Agents: Bicarbonate salts of sodium or potassium are used to release carbon dioxide in dough and batter during baking to give characteristic porous structure in bread and cakes etc.

Water retaining agents: Polyhydric alcohols (propylene glycol, PEG) exhibit water binding properties. Ethylene glycol is toxic. It also control crystallization of sugars. IM foods (intermediate moisture foods) contain moisture 15-30% to be shelf – stable and not susceptible to microbial attack in the absence of refrigeration. IM foods includes dried fruits, jams, jellies and fruit cakes.

Anticaking agents: These are added to maintain the free-flowing nature of granular and powdered food materials which are otherwise hygroscopic. They absorb excess moisture from the food (Calcium silicate and calcium stearate in baking powder and in table salt).

Firming agents: Heat processing & freezing of vegetables and fruits softens the cell walls and destroys the integrity of the products. Addition of calcium salts of chloride, citrate, lactate retain the texture of the cell wall. Ammonium alum retains the crispness of vegetables.

Clarifying agents: They prevent haze formation (cloudines), sedimentation and deterioration in wine, beer and fruit juices. Bentonite is used as clarifying agent in wine to prevent protein precipitation.

Gases and Propellants: Hydrogen is used for hydrogenation of fats. Chlorine to bleach flour, ethylene to promote ripening of fruits. Carbon dioxide for carbonation of soft drinks, beer and some wines and fruit juices to become effervescent and tangy. Nitrogen, nitrous oxide , Freon 115 are used as propellants to dispense liquid, foam and spray products.

Food Colourants

CARTENOIDS:

Cartenoids are pigments responsible for yellow and orange colour of fruits and vegetables. Chemically they are classified as terpenoids. Terpenoids having one, two, three or four isoprene units are well known; but steroids which are triterpenoids and Carotenoids, the only known tetraterpenoids, are the most important from the point of view of food chemistry. The cartenoids are of two groups, carotenes, which are hydrocarbons and xanthophylls which contain oxygen. Due to their instability and insolubility in water they are not used as food colourants. However, there are two exceptions-crocetin and bixin. Crocetin is the major pigment of the spice, saffron. Bixin is the major constituent of a colouring matter known as annatto, an extract of the fruit of the plant Bixa

orellana, a large shrub of tropical origin. Annatto is the colourant of red cheeses (Leicester cheese) and is also used to colour margarine.

Anthocyanins

Anthocyanins cause pink, red, mauve, violet and blue colour of flowers, fruits and vegetables. Cyanidin is the most important anthocyanin pigments. Grapes contain anthocyanin pigments. Anthocyanins complex metal ions giving rise to unusual colours in canned fruits. Sulphites of meta-bi-sulphites usually added as microbial preservatives in wines bleach the anthocyanin pigments, particularly in red wines. Anthocyanins also undergo a number of reactions slowly with other flavanoids present in the grapes resulting in extensive polymerization products giving a brown colour to old wines. In most food processing operations the anthocyanins are quite stable, particularly in low pH conditions.

Betalaines

Betalaines give the characteristic red-purple colour of beet-root, Beta vulgaris. These are fairly stable under food processing conditions. However in the presence of air at neutral pH, heating causes breakdown of these compounds. Betacyanins obtained from beet-root juice are used in confectionery. The beet-root juice is fermented with yeast to eliminate its high sugar content and then dried to give a powder having 6-7% of betacyanin.

Melanins

Melanins are formed in vegetables and fruits such as apples and bananas as a brown colour when the plant tissues are damaged or cut portions are exposed to air. The browning is due to polyphenolic substances of melanin type pigments produced by action of the enzyme phenolase also known as polyphenols oxidase) present naturally. During fruit and vegetable processing, the activity of phenolase is minimized by immersion in water to reduce contact with air. Phenolase necessity is inhibited by agents such as matte acid and citric acid which complex copper that is a component of active sites of the enzyme. Sliced fruits and vegetables (peaches, potatoes) are often immersed in solutions of these acids supplemented with ascorbic acid or sulphite. Ascorbic acid is not only valued as a vitamin but as a good chelating agent and antioxidant dehydrated potato powders include ascorbic acid to ensure which by preventing browning, due to traces of oxygen present in sealed cans.

The enzymatic oxidation of polyphenolic substances is a highly desirable feature of tea. The tea leaves contain about 30% by weight of polyphenolic materials. Caffeine, the stimulant in coffee and cola-based drinks, is also present in tea leaves (dried) to the extent 3-4% by weight. After the freshly plucked leaves have been allowed to wither, i.e., allowed to lose 20% of their water content, the

leaves are macerated so that the phenolase of the leaves can come into contact with polyphenolic substances of the cell vacuoles. The macerated leaves are then left at ambient temperature for a few hours to 'ferment' during which time a high proportion of the polyphenolic substances is oxidized and polymerized. The leaves are then 'fired', i.e., dried out a temperature up to 75°C, to bring the reactions to a halt and give the final product, tea.

SWEETNESS AND NON-SWEETNESS IN FOOD PROCESSING

Sweetness

Sweetness is detected by taste buds at the tip of the tongue. However, buds at the base of the tongue detect a few sweet substances (dihydrochalcones). Sweetness has been attributed to be a special characteristic of sugars. However, sugars are much less sweet than sucrose, and some varieties of sugars are not at all sweet. The molecular structure responsible for sweetness of substances has been identified as the saporous unit consisting of a proton donor/acceptor system designated as A,H-B system, A, H-B system of the sweet molecule has to match steric requirements and can interact with complementary receptor system in the taste buds through hydrogen bending, as shown. The concept has been extended to non-sugar sweet molecules also. The A, H-B system and the γ-site of a few sweet molecules also are shown in figure.

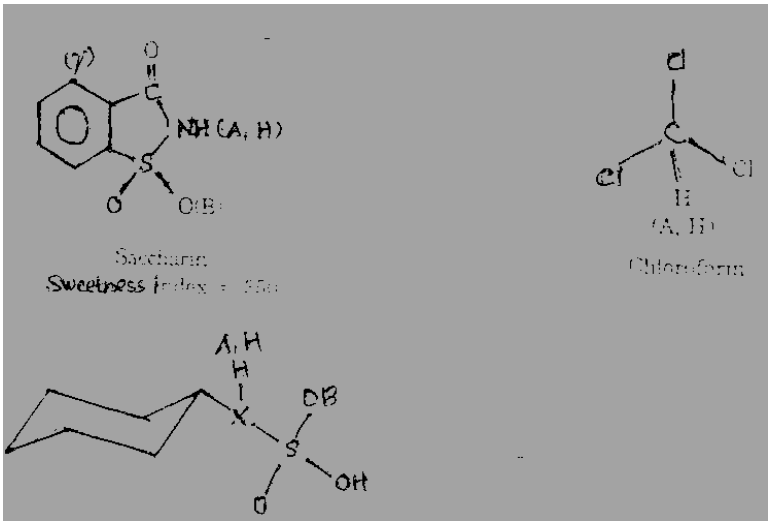


Figure: The Saporous unit and lipophilic site in a few sweet molecules

Non-sugar sweet substances

A variety of substances show sweet characteristics. The intensity of sweetness of the various substances are indicated by sweetness index (S.I.) values based on sucrose as the standard (S.I. = 1.0).

Aspartame (S.I. =200) is a dipeptide ester.

Monellin (S.I. =3000) is a sweet protein of 11.5 kDa present in the pulp of the fruit Dioscoreophyllum cumminsii. The presence of two peptide chains A dn B which are not covalently bound have been identified. The separated individual chains are not sweet and when the chains are recombines, sweet taste is restored slowly but the intensity of sweetness of native protein is not reached.

Thaumatins (S.I = 2000) are two sweet proteins (thaumatin I and II) isolated from the fruit of Thaumatococcus daniellii. The fruit also contains three other thaumatin proteins (a, b and c) in smaller amounts. Thaumatin I with 207 amino acids residues and monellin A and B peptides have been shown to have five identical tripeptide units of Thr-Arg-Gly, Ala-Asp-Ile, Asp-Ile-Ser and Gln-Tyr-Gly positively identified with sweetness of the proteins. Thr-Threonine, Arg-Arginine, Gly-Glycine, Ala-Alanine, Asp-Asparagine, Ile-Isoleucine, Ser-Serine, Gln-Glutamine, Tyr-Tyrosine.

Miraculin is a glycoprotein present in a tropical fruit known as miracle berry. It is a taste modifier because the protein has no taste but changes the taste of acidic solutions into a sweet taste. When the mouth is first rinsed with miraculin solution, lemon juice tastes sweet.

Other non-sugar sweet molecules with S.I values given in parenthese include sodium cyclamate (30), saccharia (350), neohesperidin (1000), a dihydrocahlcone derivative, perillaldehyde antioxime (2000), stevioside (300) extracted from the leaves of Stevia rebaudiana, (+)-hernandulcin (1250) a sesqui-terpene from Lippia dulcin Trev. (Verbenacceae) and 1-n propoxy-2-amino-4-nitrobenzene (4000). Sugar alcohols sarbitol and xylitol are sweet. Xylitol imparts cooling flavour in chewing gums.

FLAVOUR COMPONENTS

SALTINESS

Saltiness is detected on the sides and tip of the tongue and elicited by many inorganic salts besides common salt. Sodium chloride, particularly the sodium ion has a major role in imparting flavour to foods. For example, when sodium ions were eliminated from a mixture of amino acids, nucleotides, sugars, organic and other compounds known to mimic the flavour of crab meat, the mixture lost the flavour. Besides this tendency to stimulate meaty flavours, salt tends to decrease the sweetness of sugars, to give a richer, and more rounded flavour that is required in many confectionary products.

SOURNESS

Sourness is always assumed to be a property of acidic solutions. In most fruits and fruit juices, citric acid and malic acid are responsible for all the acidity and sourness. Tartaric acid is an essential characteristic of grapes while isocitric acid is that of black berries. Oxalic acid is present in rhubarb. The individual content of organic acids varies in fruit juices. For example, orange juice citric acid (51 mM) and malic acid (13 mM), while lemon juice contains higher concentration of citric acid (220 mM) together with smaller amounts of malic (7 mM) and citric acid (16 mM).

BITTERNESS

Bitterness is associated with several classes of chemicals and it is detected by taste buds at the back of the tongue. Structural criterion (size) distinguishes saltiness from bitterness in the case of alkali metal halides. When the sum of the ionic diameters is below that of KBr (0.658 nm) which tastes both salty and bitter, then the salty taste predominates as in NaCl (0.556 nm). Kl (0.706 nm) and $MgCl_2$ (0.85 nm) are bitter Phenolic substances in the form of flavanoids are important sources of bitterness in fruit juices, particularly citrus fruits, Naringin, a glycoside of the falvanone naringenin, occurs in grape fruit and Seville oranges. Its bitterness can be detected even at a dilution of 1 in 50,000. Limonin is another bitter substance in citrus juices.

Bitterness is preferred in beer. Bitterness is achieved by adding hops to the wort (unfermented sugary extract of the malt. The mixture is boiled, cooled and allowed to undergo fermentation. Hops are dried flowers of the hop plant Humulus lupulus, rich in volatile compounds and these give been its characteristic aroma together with resins and α-acids which give bitterness to beer.

ASTRINGENCY

Astringency is a sensation related to bitterness but is registered within the oral cavity as well as on the tongue. It is a desirable characteristic of fruits and cider but is only in red wine and tea that astringency is most important. It is associated with the high content of polyphenolic compounds which are also involved in the production of colour. In tea and wine the polyphenols are called fennins.

PUNGENCY

Pungency is another sensation experienced by the entire oral cavity and an essential characteristic of spices such as green and red chillies, pepper, ginger, mustard, onions, garlic and to a lesser extent, in leafy vegetables (cabbage).

MEATINESS

The flavour of a meat is an amalgamation of our responses to both volatile and non-volatile compounds. Meaty taste is associated with two substances which occur in many other foods besides meat. They are ionosine monophosphate (IMP) and monosodium L-glutamate (MSG). They are known as flavour potentiators or modifiers because they have no flavour of their own, but can enhance desirable flavours. A concentration of 300 mg/dm^3 is necessary to make an impact. When present together they show synergism. A mixture of equal proportions of IMP and MSG is about 20 times stronger than the same total amount of one alone. MSG is used as flavour enhancer in dried soups, at 1 g/dm^3 concentrations.

NATURAL FLAVOUR DEVELOPMENT IN FOODS

The interest in developing natural food flavours has paved way for biotechnological processes. Two broad strategies are adopted for flavour development in foods. Microorganisms or enzymes are used to generate flavouring complexes or multi-component flavour systems as practiced in the production of dairy products. Alternatively, biosynthesis may be facilitated to develop single-flavour compounds.

Multi-components flavour systems are developed by the use of specific trains of lactic acid bacteria and fungi to arrive at specific cheese flavours. For example, the specific flavour of blue cheese is developed by Penicillium sp., mainly P. roqueforti which converts the free fatty acids formed by lipolytic reactions in cheese to methylketones are formed in two-step process of β-oxidation of the fatty acids to β-keto acids which undergo decarboxylation to give methylketones of which the main products include 2-pentamone, 2-heptanone and 2-nonanone. Another example of the development of multi-component flavour system involves the use of short-chain fatty acid specific lipases and proteases for the production of enzyme modified cheeses. Other examples are included in section.

Biosynthesis and isolation of individual flavour compounds is the second strategy involving biotransformation through hydration/dehydration or oxidation/reduction reactions. For example, the buttery flavour of fermented dairy products such as sour cream and buttermilk is primarily due to diacetyl, biosynthesized from citric acid naturally present in milk. The pleasant coconut and peach like fruity flavours natural to fruit, vegetables, nuts and milk products are due to lactones such as γ-decalactone formed by yeast species (Candida sp) and fungi (Sporobolonyces odorous, Trichoderma viride) using ricinoleic acid (12-hydroxyoctadecyl-9-enoic acid) as the substrate.

Biosynthesis of esters such as ethyl acetate, ethyl butyrate, ethyl isovalerate and ethyl hexenoate responsible for the fruity aroma and flavour is facilitated by the use of species of Pseudomonas, Luctococcus, Lactobacillus and Candida utilis. Pyrazines and their derivatives are associated with

roasted and nutty flavours (found naturally in green beans and potatoes) and are biosynthesized by species of Lactococcus lactis, Pseudomonas, perotens, Bacillus subtilis and Corynebacterium glutamecum. The pungent taste of mustard, horseradish and cress is due to the formation of isothiocyanates from glucosinolates by the action of the enzyme myrosinase. The flavour and aroma of cucumber, apple and tomato are identified with green flavour components such as nonenal, nonadienal and hexanal. These compounds are formed by degradation of unsaturated fatty acids via a lipoxygenase catalyzed formation of hydroperoxides followed by cleavage by a hydroperoxide lyase.

Flavour enhancers such as inosine-5' monophosphate (IMP) and guanosine-5' monophosphate (GMP) can be produced by degradation of RNA with 5'-phosphodiesterase. RNA is a by-product of plant processing and single cell protein (SCP) production.

Wood Smokes Functions as a Preservative and Flavour Enhancer

The smoking of foods is practiced mainly for two purposes: adding desired flavour and preserving food. Other desirable effects may also be obtained, e.g., improvement in the colour of inside of meat and in the finish or gloss of the outer surface and a tenderizing action on meats. The smoking process helps preservation by impremating the food near its surface with chemical preservatives from the smoke. The combined action of heat, preservatives and drying at the surface is responsible for the preservation action.

Smoke is obtained from the burning wood preferably wood like hickory, but it may be generated from burning corn cobs or other materials. Oak, maple, beech, birch, walnut and mahogany wood are also used particularly for apples. Sawdust is added to the fire to give a heavy smudge. Temperature and humidity are controlled to remain at certain levels favourable to the product being smoked and the duration of smoking depends upon the kind of food. Smoking temperatures for meats vary from 43 to 71°C and the period lasts from a few hours to several days.

Wood smoke consists of two phases, a dispense phase of liquid droplets and a continuous gas phase. The gas phase of smoke contains over 200 different volatile compounds that have bacteriostatic and bactericidal effect. Formaldehyde is the most effective of these compounds. Other compounds include phenols, cresols, aliphatic and from formic to caproic acids, primary and secondary alcohols, ketones, acetaldehyde and other aldehydes, waxes, resins, guaiacol and its methyl and propyl isomers, catechol, methylcatechol and pyrogallol and methyl ester. Smoke also contains carcinogens such as benzpyrenes and other polynuclear hydrocarbons. These various compounds of wood smoke are grouped under the name pyroligneous acid. Wood smoke is more effective against vegetative cells than against bacterial spores and the rate of germicidal action of the

smoke increases with its concentration and the temperature and varies with the kind of wood employed. The application of liquid smoke (solution of chemicals similar to those in wood smoke) to the outer surface of foods has little or no preservative effect although it contributes to flavour.

FLAVOUR ENHANCERS

These compounds enhance the aroma of a food apparent to the senses as 'feeling', 'volume', 'body', or 'freshness' particularly in thermally processed foods. Monosodium glutamate (MSG) gas a pleasant, salty-sweet taste and a property often described as 'mouth satisfaction'. It promotes sensory perception of meat-like aroma and is used as additive in frozen, dehydrated or canned fish or meat products. Inosine monophosphate (IMP) and guanosine monophosphate (GMP) have properties similar to that of MSG and find use in soups, sauces, canned meat or tomato juice at 75-100 ppm levels. Maltol (3-hydroxy-2-methyl-4-pyrene) has a caramel like odour and enhances the perception of sweetness of carbohydrate rich foods such as fruit juices, marmalades and jelly. Addition of maltol at 5-75 ppm allows a decrease of sugar content by about 15% without affecting the sweetness intensity.

Enzymes in Food Processing

PECTIC ENZYMES

Pectin is polygalactouronic acid with about two-thirds of carboxyl groups existing as methyl ester. Microbial spoilage of fruits and vegetables is primarily due to the action of pectic enzymes of pathogenic organisms which facilitate the invasion of tissues. Native or endogenous pectic enzymes present in foods are also involved in the ripening as well as in the spoilage of plant tissues. Pectic enzymes such as pecting methyl esterase (PE), polymethylgalactouranase (PMG), polygalactournase (PG) and pectic acid transeliminase (PATE) are used for clarifying fruit juices, in controlling the extent of ripening of fruits, maintenance of viscosity in fruit products and in achieving textural changes. In jam manufacture the pectin esterase activity is inhibited by high concentration of sugar and thus the liquefaction of pectin gel is prevented.

AMYLASES

Two amylases, α and β-amylases are well known for the hydrolysis of starch during ripening of fruits, processing of potatoes, production of corn syrup and corn sugar and in bread making. α-1, 4-glycosidic linkages at random and hence used as 'liquefying enzyme' for reducing the viscosity of starch solutions. Starch hydrolysis by the use of this enzyme also increases reducing sugar content of processed foods. β-Amylase is an exoenzyme, occurring in vegetable tissues, which removes the maltose units from non-reducing end of the starch chain by hydrolyzing alternate glycosidic linkages. Since maltose increases the sweetness of the starch solution, β-Amylase is

known as 'saccharifying enzyme' β-Amylase activity stops at two or three glucose units from branching point of amylopectins resulting in incomplete hydrolysis of starch to give 'limit dextrins'.

CATHEPSINS

These are intracellular acidic proteases located in the lysosomal fraction of animal tissues and are distinct from proteases such as trypsin and chymotrypsin secreated by cells. Five cathespins designated as cathespins A, B, C, D and E are known and may be involved in the tenderization of meat during aging.

MYOSIN ATPASE

The enzyme plays an important role in muscular contraction in living systems and also in the rigidization of muscle due to extreme contraction and shortening during postmortem rigor mortis.

FLAVOUR ENZYMES

Enzymes participate either directly or indirectly in the formation of flavour compounds in fruits and vegetables. Four types of flavour forming mechanisms have been identified in foods and three of these mechanisms require enzymes directly or indirectly.

1. Biosynthesis. Flavour compounds such as terpenoids, esters alcohols and acids are formed enzymically during the ripening of fruits such as apples, bananas, pears and citrus fruits and in pepper (these plant foods do not have any flavour during initial stages of growth up to harvest).
2. Direct enzymic reactions. A specific flavour compound is formed by the direct action of an enzyme on a specific precursor. For example, the flavour of onion is due to the action of endogenous S-alkyl-L-cysteine sulphoxide lyase on the trans(+) S-propenyl-L-cysteinesulphoxide and other S-alkyl substituted L-cysteines naturally present in the onion to give volatile compounds with characteristic pungent aroma and taste of onions. A part of flavour precursors in onions is present in the form of γL-glutamyl peptides which can be released by the addition of exogenous γLg-glutamyl transpeptidases so that full potential of onion flavour may be achieved. Another example of direct enzymic action on flavour precursors is that of cabbage flavour in cabbage, mustard and horseradish. Thioglycosidases act on thioglycoside precursors such as sinigrin or sinalbin to give alkylthiocyanates in cabbage flavour.
3. Oxidative or indirect enzymic reaction. The mechanism involves formation of the flavour components by oxidation of flavour precursors, the oxidizing agents being formed enzymically. In the black tea aroma, catechol oxidase oxidizes flavonols and the oxidized flavonols in turn oxidize amino acids, unsaturated fatty acids and amotenes present in tea to yield flavour constituents. Lipoxygenase present in plants, form hydroperoxides which are capable of oxidizing

susceptible food constituents to give flavour compounds.

4. Pyrolysis. Pyrolytic mechanism does not involve any enzyme. Flavours are produced by the action of heat on precursors during heat processing or baking. Examples include the flavours of coffee, chocolate and bread.

The use of enzymes to regenerate flavour in processed foods is known are flavourase concept. During heat processing native flavourase enzymes are inactivated and most of the flavours volatilize off. Exogenous enzymes from a homologous source when added to processed foods can act on flavour precursors which are non volatile to regenerate flavours. Examples of processed foods in which flavour has been partially or fully regenerated include cabbage, potato, tomato, peas, carrots, bananas, grapefruit juice, orange juice, mustard, horseradish and onions.

LIPOLYTIC ENZYMES

Lipases hydrolyse ester linkages of emulsified glycerides at oil-water interface. These enzymes are present in cells, seeds, organs, tissues and fluids of plants and animals and in microorganisms. Generally lipases hydrolyze triglycerides in a stepwise manner to 1,2 or 2,3-diglycerides and then 2-monoglycerides. Lipases show four types of specificities glyceride, position, fatty acid and stereo specificities. Action of endogenous lipases on triglycerides can lead to off flavours in foods. The bovine milk contains the lipase which is normally inactive but homogenization and cooling during milk processing activate the enzyme. This induced lipolysis results in the formation of short and medium chain length volatile fatty acids and spoilage of milk.

LIPOXYGENASE

Lipoxygenase or lipoxidase is present in legumes such as soybeans, mung beans and green beans, in cereals such as wheat, corn, oats and barley and also in fruits. The enzyme catalyzes stereospecific oxygenation of polyunsaturated fatty acids. Esters and glycerides containing the cis, cis1,4-pentadiene group to give optically active hydroperoxide containing conjugated cis, trans double bond system. Thus linoleic and linolenic acids present in most vegetables are subjected to peroxidation by lipoxygenase resulting in the formation of off-flavours. The enzyme also bleaches pigments such as chlorophyll, carotene and xanthophylls. Because of the formation of free radicals during the action of this enzyme, vitamins are destroyed. Inactivating the enzyme by blanching vegetables before freezing or drying may prevent these undesirable effects. However, the enzyme is used beneficially in bleaching and improving the dough characteristics of wheat flour used for bread making by way of addition of soya flour. The addition of soya flour decreases the content of

polyunsaturated fatty acids and acids in the release of bound lipids. The improvement of dough occurs due to lipid oxidation during mixing in the presence of oxygen and also possible due to the oxidation of sulphydryl groups of proteins to give disulphide bridges resulting in crosslinked proteins.

PEROXIDASE

Milk and horseradish are best known sources of this enzyme. The enzyme catalyzes the reduction of peroxide group by electron donors such as ascorbate, phenols, amines or other organic compounds. The oxidation product is usually highly coloured and has been used for the colorimetric analysis of peroxidase activity. Enzymic analysis of foods uses these colour reactions. For example, in the estimation of glucose by glucose oxidase, the by-product is hydrogen peroxide which can be estimated with peroxidase reaction, and from the known stoichiometry of glucose oxidase reaction glucose can be estimated. Since peroxidase is very resistant to heat, it can serve as an indicator of the effect of heat treatment of a food such as pasteurization of milk. Peroxidases are involved in the destruction of Vitamin C, bleaching of Carotenoids and anthocyanins and also in the peroxidative degradation of fatty acids to give volatile carbonyl compounds responsible for oxidized flavour.

THIAMINASE

Thiaminase is present in most fresh water and marine fish and in unpolished rice, beans and mustard seed. The enzyme hydrolyzes thiamine and causes thiamine deficiency diseases such as beriberi. Thiaminas activity is destroyed by cooking or boiling fish. Salting the fish also reduces the enzyme activity.

ASCORBIC ACID OXIDASE

In the absence of enzyme, Vitamin C is oxidized in the presence of air by copper or ferric ions to give dehydron carbic acid and hydrogen peroxide. The hydrogen peroxide formed destroys ascorbic acid. Ascorbic acid oxidase is a copper containing enzyme which catalyzes the oxidation of ascorbic acid to dehydroascorbic acid and water. The enzyme is present naturally in oranges, citrus fruits and vegetables but not found in animals. The presence of reductases such as glutathione dehydrogenase in the whole fruit balances the activity of ascorbic acid oxidase by reducing dehydroascorbate. However, during extraction of fruit juices, reductases suffer maximal damage and ascorbic acid oxidase by reducing the whole fruit balances the activity of ascorbic acid oxidase activity increases. Deaerating the juice to remove oxygen and also pasteurizing it subsequently

inhibites the activity of the oxygen and also pasteurizing manufacture, the orange peals are boiled in water to reduce the activity of the enzyme. Other oxidases such cytochrome oxidase peroxides and phenolase also catalyze the destruction of ascorbic acid.

ENZYMES USED AS FOOD PROCESSING AIDS

Enzymes catalyze a variety of reactions during processing and preservation of food products and form a large group of processing aids. Food industry utilizes enzymes in the production of bakery items, dairy products, fruits and vegetables juices and products, tenderization of meat, modification of oils and fats, flavour production and preservation of food products. Enzymes preparations may be added during processing or the process conditions may be added during processing or the process conditions may be manipulated to make use of native or endogenous enzymes.

BREAD MAKING

Wheat flour, water, salt and yeast are basic ingredients of bread and other baked goods. A variety of additives are included to improve the flour quality, dough making and handling, baking characteristics and quality of the finished product. Endogenous proteases and proteases preparations from Aspergillus oryzae have been used as processing aids in improving the quality of flour to achieve desired elastic characteristic of the dough. The high molecular weight glutten proteins initially break down due to the combined effect of hydration, mechanical mixing as well as the hydrolytic activity of proteases. Later, during fermentation the gluten proteins form a cohesive network through disulphide bridges and entrap carbon dioxide, thereby making the dough elastic. The formation of non-elastic 'bucky' dough has been attributed to the higher content of high molecular weight of gluten proteins in the flour which require longer mixing times. Proteases are added particularly to flours containing higher gluten content to hydrolyze the high molecular weight glutens partially and thereby enhance the elastic characteristics of the dough.

Endogenous redox enzymes such as peroxides, ascorbate oxidase, dehydroascorbate reductase glutathione oxidase and grahlone reductase along with chemicals such as bromate or ascorbic acid promote the disulphide bride formation. Other redox enzymes such as glucose oxidase, polyphenol oxidase and peroxidase may be added to the flour though such addition has not been practiced commercially.

Lipoxygenase present in the soya flour added to wheat flour for bread making promotes aggregation of gluten proteins acid also catalyzes the oxidation of poly unsaturated fatty acids forming acids. The peroxides are strong oxidizing agents and bleach the caroteacid pigments thereby improving the colour of the wheat flour.

Wheat flour has relatively less endogenous α-Amylase acid the enzyme is supplemented by the addition of fungal α-Amylase to enhance the rate of starch degradation in order to maintain yeast fermentation and obtain satisfactory loaf volume. Starch is hydrolyzed to produced limit dextrins which are hydrolyzed by endogenous β-Amylase to fermentable glucose. Alternatively, fungal glucoamylase may be used to release glucose from starch. Thus the combination of protease, oxidases and glucoamylase improve the leavening of bread.

During baking, gelatinization of starch occurs during which the starch granules swell by absorbing water, lose their crystalline structure and exude soluble amylase resulting in a concentrated starch suspension which is transformed into a starch gel. The strength of this gel depends on the amylase concentration. Recrystallization of amylase during baking and cooling results in the formation of a rigid three-dimensional network and fixation of crumb structure. A fast crumb setting reduces the 'ovenspring' giving rise to a small loaf volume. The activity of α-amylase, which continues during baking till the internal dough temperature main polysaccharide in the middle lamelia of the cell wall. The enzyme from plants acts preferentially on methoxyl groups of esterifies dgalacturonic acid at the non-reducing end of the polygalacturonan chain in a sequence while pectin esterases of fungal origin hydrolyze the methoxyl groups in a random manner and hence decreases the viscosity of the pectin solution.

UNIT III

MICROORGANISMS ASSOCIATED WITH FOOD

RELEVANCE OF MICROBIOLOGY IN FOOD PROCESSING

Food is an interacting medium between various living species because it is a source of nutrients for humans, animals, and microbes. Food fit for human consumption is also a medium for the growth and activity of microbes. Hence human food is always associated with a variety of microbes. It use human food as a source for itsown growth and activity. Microbial activity in food can be beneficial in cartain cases but in most of the cases it leads to deterioration of the food and renders it unfit for human consumtion. Four aspects of microbiology relevant to food processing and preservation are

i) Fermented foods: microorganisms can be used as processing aids in the production of fermented foods. New and modified foods with better shelf life, palatability, flavor and organoleptic properties are produced by fermentation using specific microorganisms under controlled conditions.

ii) Food chemicals from microorganisms: A variety of food chemicals and additives may be produced by the fermentation using specific microorganisms. Sometimes microbes itself can be used as food. The biomass produced by fermentation can be harvested and used as protein rich raw material for the formulation of foods

iii) Food poisoning and food borne diseases: pathogenic microbes grow in the food utilizing nutrients in the food and produce toxins which are detrimental to the health of the consumer when such food is consumed. Food also serves as a vector or medium for certain pathogens that cause food infection and diseases.

iv) Food spoilage: The metabolic activity of various microorganisms not only utilizes the nutrients in food but also causes the spoilage of food through undesirable enzymatic changes affecting the quality of the food. The enzymatic changes include the formation of products which contribute off-flavours and affect the organoleptic , textural and keeping qualities of the food.

FACTORS AFFECTING MICROBIAL ACTIVITY

The Intrinsic factors affecting are

- pH value of the food
- Moisture content
- Water activity
- Oxidation –reduction potential
- Nutrient content
- Presence of inhibitory substances.

The extrinsic factors affecting core of the following

- Temperature
- Relative humidity of environment
- Gaseous atmosphere surrounding the food

Factors influencing microbial activity

Growth and activity of microbes depend on the nature of food and its composition. The factors that govern the microbial activity in a food include both intrinsic and extrinsic factors. Intrinsic factors include pH, water activity, oxidation reduction potential, nutrient content and presence of inhibitors in the food. Extrinsic factors include temperature, relative humidity and atmosphere surrounding the food.

A. Intrinsic factors

1. pH: Plant and animal food have an natural pH in the range of about 2 to neutral pH of about 7. Every microorganism has an optimum pH at which its growth is maximum and has minimum and maximum pH values it can tolerate. Yeast and molds are more tolerant to lower pH compared to bacteria. Foods with relatively acidic pH of less than 4.5 are not easily spoiled by bacteria. Thus fruits have pH values below those required by many spoilage organisms. The keeping quality of meat from fatigued animal is low compared to that of well rested animals because of pH attained upon completion of rigor mortis. Some foods have buffering capacity in that they resist changes in pH. In general buffering capacity of meat is greater because of higher protein content compared to vegetables which lack buffering capacity.

2. Moisture content and water activity: Water requirements of microbes are generally expressed in terms of water activity of the substrate in which they grow. Water activity of most fresh foods is in the range of 0.9 to 0.99 and such an environment is highly conducive to growth and activity of spoilage microorganisms. Many preservation methods are oriented towards decreasing in the water activity in foods.
3. Oxidation reduction potential: O-R potential of a substrate is defined as ease with which substrate loses or gains electrons and is expressed by the symbol Eh. When electrons are transferred from one compound to another a potential difference exists between the two compounds which is expressed in millivolts. Fruit juices are highly oxidised substrates with Eh values in the range of +400mV. Solid meat is reducing medium with Eh values in the range of -200mV while minced meat is an oxidising medium with Eh values in the range of +200mV. Aerobic microbes require positive Eh values for their growth while anerobic organisms require negative Eh values. The O-R potential of a food is determined by
   a) Characteristic O-R potential of the original food
   b) Poising capacity-ability to resist change in potential of the food
   c) Oxygen tension of the atmosphere surrounding the food
   d) Access of atmosphere to the food

4. Nutrient content: microbes require water, source of energy, source of nitrogen, vitamins, minerals and other growth factors. Most microbes use carbohydrates, especially fermentable sugars and other carbon compounds. Most microbes are capable of using only glucose, they differ in their ability to use various nitrogenous compounds as their source of nitrogen.
5. **The presence of inhibitory substances:** Anti microbial substances such as spices have essential oils which provide anti microbial characteristics. Milk contains anti microbial constituents lactoferris, corn glutinin and lacto peroxidise. Eggs contain lysozyme and corn albumin. Anti microbial may also be added deliberately during

B. Extrinsic factors:

1. Temperature: Microbes grow over a wide range of temperatures, the lowest temperature at which the microbes grow is about -34°C and the highest temperature is about 90°C. Organisms that grow below 20°C and have optimum temperature in the range of 20°C to 30°C are referred to as psychrophiles. Organisms that grow well between 20°C and 45°C with optimum temperature in the range of 30 and 40°C are called mesophiles. Thermophiles are those organisms that grow well are temperatures above 45°C with optimum temperature in the range of 55°C to 65°C. The type and quality of the food must be taken into account for selecting a storage temperature. For example, bananas keep better at 13°C to 17°C while a large number of vegetables store better at about 10°C.
2. Relative humidity of environment: The relative humidity of the storage environment must be such that excessive drying of the food or absorption of moisture by the food does not occur.
3. Gaseous atmosphere surrounding the food: It is possible to inhibit surface spoilage of foods by a controlled atmosphere surrounding the food. Increased amounts of carbon dioxide in the atmosphere by about 10% retards fungal rotting of fruits while higher percentage of carbon dioxide is effective with meat, fish and other animal products. Addition of ozone has been found to be effective against several spoilage microorganisms. Ethylene has been found to be effective in maintaining the natural resistance of frits against fungal attack

IMPORTANCE OF BACTERIA IN FOODS

Bacteria that play significant roles in foods are often grouped on the basis of their activity in foods without regard to their systematic classification.

Lactic acid bacteria ferment sugars to lactic acid and include species belonging to genera of Leuconostoc, Lactobacillus, Streptococcus and Pediococcus. Their activity is

desirable in a variety of foods such as sauerkraut and other pickled vegetable and dairy products for the production of flavour. They cause spoilage of wines.

Acetic acid bacteria oxidize ethanol to acetic acid. Species of genera Acetobacter and Gluconobacter are the most common. Acetobacter species are capable of oxidizing acetic acid to carbon dioxide. They are useful in vinegar manufacture but are undesirable in alcoholic beverages.

Butyric acid bacteria are mostly the spore forming anaerobes of the genus Clostridium. They produce butyric acid by fermenting sugars. Propionic acid bacteria produce propionic acid and belong the genus Propionibacterium.

Proteolytic bacteria include aheterogenous group of bacteria which produce extracellular proteases. The species include aerobic, facultative and spore forming organisms. Most species belong to the genera of Clostridium, Bacillus, Pseudomonas and Proteus. Bacillus cereus in an aerobic, spore forming organism while pseudomonas fluorescens is non spore forming, aerobic as well as facultative organism.

Lipolytic bacteria are also a heterogenous group of bacteria which produce lipases. Many of the aerobic proteolytic bacteria are also lipolytic. Organisms of the genera Pseudomonas, Alcaligenes, Staphylococcus, Serratia and Micrococcus are lipolytic.

Saccharolytic bacteria hydrolyze disaccharides and polysaccharides to simple sugars. Bacillus subtilis and Clostridium butyicum are examples of saccharolytic bacteria which are also amyloytic, ie., produce amylase to hydrolyze starch outside the cell.

Pectinolytic bacteria produce pectinases responsible for softening of plant tissues or loss of gelling power in various plant foods. Species belonging to genera of Bacillus, Archormobacter, Aeromonas, Arthrobacter and Flavobacterium are pectinolytic.

Thermophilic and thermoduric bacteria are resistance to high temperature. Thermophilic bacteria cause spoilage of low acid canned foods. Important species include Bacillus stearothermophilus and C. Thermosaccharolyticum. Thermoduric organisms survive heat treatment such as pasteurization. Bacteria of genera Bacillus, Clostridium, Micrococcus, Streptococcus, Lactobacillus and Microbacterium are found in foods.

Psychrotrophic bacteria are able to survive and grow at refrigeration temperature though their optimum temperatue of growth is around 20 to 30° C. In contrast, psychrophilic bacteria have optimum temperature of growth at refrigeraton temperature. Psychrotrophic bacteria spoil nor-sterile refrigerated foods such as meat, fish, poultry, eggs and other foods. Species belonging to genera of Psuedomonas, Achromobacter, Alcaligenes and Flavobacterium are psychrotrophic.

Halophilic bacteria include species of the genera Bacilus, Micrococcus, Vibrio, Maraxela, Halobacterium, Corynebacterium, Strptococcus and Clostridium. These organisms require certain minimal concentrations of dissolved sodium chloride for their growth and survive at higher concentrations of salt to different extents depending on whether they are slightly, moderately or extremely halophilie or halotolerant species.

Osmophilic or saccharophilic bacteria grow in high concentrations of sugar, e.g. Leuconostoc sp.

Pigmented bacteria produce colour during their growth in foods. Examples include species belonging to genera Flavobacterium (yellow to orange), Serratia (red), Halococcus (red to orange)and Halobacterium (pink, red and organge).Lactobacillus plantarum produce rust colour pigment discolouring cheese. Flavobacterium species cause discolouration on the surface of meat and spoilage of shellfish, poultry, eggs, butter and milk.

Slime or rope forming bacteria include Alcoligenes viscolactis, Enterobacter aerogenes, and Klebsiella oxytoca and some species of Streptococcus and Lactobacillus which cause slime or ropiness in milk. Lactobaeillus plantarum causes ropines in various fruit, vegetable and grain products such as cider, sauerkraut and beer while some species of the genus Bacillus cause ropiness in bread. Meat curing solutions become slimy due to the activity of Micrococcus. Gluconobacter species includes G. oxydans, which causes ropiness in beer.

Gas forming bacteria include species of the genera of Leuconostoc, Lactobacillus and Propionibacterium which produce carbon dioxide. Species of Escherichia,

Enterobacter, Proteus, Bacillus and Clostridium produce both carbon dioxideand hydrogen, clostridium species ferment carbohydrates with the production of butyric and other acids and gases (carbondioxide and hydrogen). The species are found commonly in soils and involve in the gaseous spilage of canned vegetables and putrefaction of foods. C. perfringens causes stormy fermentation of milk.

Off-flavour forming bacteria include those of genus STreptomyces which produce undesirable flavours and musty or earth odour and taste. Many species of Pseudomonas produce a variety of metabolitics that affect the flavour of foods deleteriously.

Coliform bacteria are aerobic and facultatively anaerobic non-spore forming bacteria which ferment lactose with gas formation. They cause spoilage of a variety of foods producing off- flavours and sliminess. The important species include Escherichia coil and Enterobacter aerogenes. E. coli counts in foods are used as indicators of fecal pollution.

Several strains of the genus Bacillus are important as test organisms in sterility testing of food products. B. Subtilis and B. Stearothermophilus are recommended for testing steam sterilization procedures. B. subtilis is also the test organism for detecting penicillin in milk. B. pumilus is recommended for determining the suitability of gamma radiation sterilization. B subtilis var. niger is recommended for ethylene oxide sterilization testing.

Some of the important disease causing bacteria include the following.

C.botulinum produces a neurotoxing in canned meat products and causes the fatal disease botulism.

Corynebacterium species includes the diphtheria organism C. diptheriae.

Erwinia species are plant pathogens and damage plants and plant products causing baterial soft rot.

E. Coli species includes some serotypes which are pathogenic to humans.

Mycobacterium species includes the tubercle bacilli M. tuberculosis that causes tuberculosis especially through raw milk from infected cows.

Salmonella species are enteric pathogens that grow in food and cause food infection.

Staphylococcus species includes the important. S. cureus which produce an enterotoxin causing food poisoning.

Streptococcus species includes pathogenic S. agalactiae which causes mastitis in cows and S.Pyrogenes which causes septic sore throat, scarlet fever and other diseases in humans.

Vibrio species is pathogenic to humans.

IMPORTANCE OF YEAST IN FOOD

Saccharomyces species are the most widely used yeasts. The leading species S. Cerevisiae is used in the manufacture of many foods, with special strains used for the leavening of bread and for the production of ale, wine, alcohol, glycerol and invertase. Top yeasts are active fermenters and grow rapidly at 20°C. The clumping of the cells and the rapid evolution of carbon dioxide sweep the cells to the surface and hence the name top yeasts. Bottom yeasts do not clump and settle to the bottom, hence the name bottom yeasts. They grow more slowly and ferment at lower temperatures of 10-15°C.

S. cerevisiae var ellipsoideus is a high alcohol yielding variety use din the production of industrial alcohol, wine and distilled liquors. S.uvaraum is a bottom yeast used in beer manufacture. S. Fragilis and S. lactis have the ability, to ferment lactose and hence are important in milk products.

Zygosaccharomyces species are osmophilic and are involved in the spilage of honey, syrups and molasses. They find use in the fermentation of soya sauce and some wines.

Candida species spoil foods high in acid and salt content. C.utilis is grown for food and C. Krusei is used with dairy starter cultures to maintain the activity and longevity of lactic acid bacteria. C. lipolytica spoils butter and oleomargarine.

Brettanomyces species are involved in the fermentation of Belgian and English beers and in French wines, while Kloechera species are found commonly on fruits and flowers and in soil. Trichosporon species (T. pullanlants) grow at low temperatures and are found in breweries and on chilled beef.

Rhodotorula species cause discolouration of foods by forming coloured spots on meat and sauerkraut.

IMPORTANCE OF MOLDS IN FOODS

Even though Mucor species are involved in the spoilage of some foods, they find use in the manufacture of certain foods such as Oriental foods and ripening of cheese (e.g. Gammelost). Rouxii is used for the saccharification of starch. Rhizopus stolonifer, the so-called bread mold is a common food spoilage mold involved in the spoilage of fruits, vegetables, berries and bread. Thamnidium elegans spoils refrigerated meat by forming whiskers.

Aspergillus is a widespread species. Some of the Aspergilli are involved in spoilage of foods, e.g., A glaucus and A. repens. The widespread A. niger species is used for the manufacture o citric and fluconic acids and also a variety of enzyme preparations. A flavus-oryzae group includes molds useful in the making of Oriented foods and production of enzymes. Some molds of this group are also spoilage organisms.

Penicilium genus is widespread in occurrence and some of the species cause spoilage of foods while certain others are useful. P. expansum causes soft rots of fruits. P. digitatum and P. italicum cause rotting of citrus fruits. P. camemberti and P. roqueforti are used in ripening of cheeses. Other molds belonging to genera such as Trichothecium, Botrytis (causes gray mold belonging to genera such as Trichothccium, Botrytis (causes gray mold on many plants and plant food), Trichoderma, Cladosporium (produce black spots on beef. Endomyces, Fusarium (causes neck rot of banana and spoilage of other fruits and vegetables) and Sclerotinia cause spoilage of foods.

FERMENTATION

Fermentation process mainly divided into 3 steps

1)Upstream

2)Fermentation

3)Downstream

In upstream- strain selection, isolation,preservation, media preparation, innoculum preparation

In downstream process- filtration, cell distruption,protein purification, lyophilization and packing

FERMENTED MILK PRODUCTS

Milk is a highly unstable food as it is a very good medium for the growth of many microorganisms. Fermentation of milk is primarily aimed at preservation by conversion into more stable, nutritious and desirable products such as cheese, yoghurt, butter milk and butter. Milk from cows, sheep, goats and other mammals may be used directly for fermentation or after preliminary treatment such as pasteurization, ultrafiltration or lactose hydrolysis. The fermentation may be initiated spontaneously by native microflora or by addition of specific starter cultures or material from previous fermentation. The fermented milk may undergo further processing to yield a more concentrated product or may be blended with other ingredients such as salts, fruits, herbs, spices, sugars and natural colourants to change flavour, appearance and texture of such products.

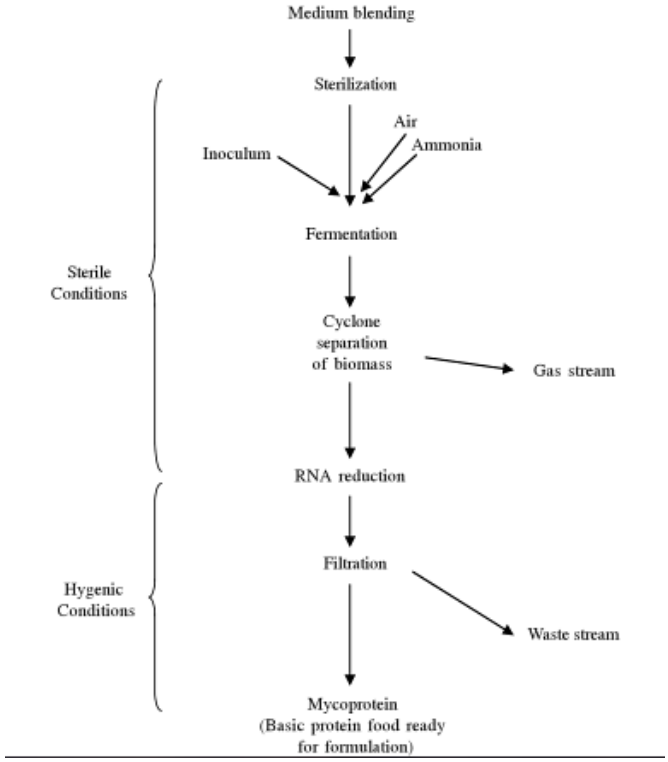
The microorganisms used in the industrial scale of fermentation of cow's mild include mainly lactic acid producing bacteria such as the mesophilic *Lactococcus* and *Leuconostoc* species with optimum growth temperatures in the range of 20 to 30°C and thermophilic *Lactobacillus* and *Streptococcus* species used at temperatures up to 45°C. Other bacteria such as *Corynebacterium, Propionibacterium,* yeasts such as *Saccharomyces* and molds such *Pencillium camemberti* and *Pencillium candidum* are also used.

The nutrients in cow's milk include lactose (~4.5%), protein (~3.5%), fat (~4%) and citrate which are utilized by the microorganisms. Lactose is converted into lactic acid homolactically or into lactic acid, carbon dioxide and ethanol heterolactically. The milk proteins are hydrolyzed to individual amino acids by the proteolytic enzymes of the starter culture. The amino acids contribute to the flavours either by themselves or by yielding aldehydes, amines, alcohols and various sulphur compounds via dicarboxylation, tansamination, deamination and desulphurization reactions. For example, *Propionibacterium* produces proline is considered to be an important contributor to the sweetness in swiss type of cheese. Another protein metabolic product is ammonia,

responsible for the flavoud of Camembert cheese. *Pencillium sp.* Produces hydrogen sulphide, dimethyl sulphide and methane thiol from methionine. The flavour of yoghurt is due to acetaldehyde derived from the degradation of treonine catalyzed by threonine aldolase of Lb.bulgaricus. Citrate is metabolised by Lb. Lactis, L.cremoris and others to an intermediate product α-acetolactate, which is decarboxylated to diacetyl, the characteristic flavour compound of butter, buttermilk and cottage cheese. In addition, carbon dioxide is also an end product of citrate metabolism contributing to the formation of 'eyes' in certain types of cheese such as Gouda cheese. The milk fat is not utilised directly by starter cultures of lactic acid bacteria as they are weakly lipolytic. But other organisms constituting the secondary flora hydrolyze milk fat to liberate free fatty acids.

SINGLE CELL PROTEIN

Single-cell protein typically, referring to sources of mixed protein extracted from pure or mixed cultures of algae, yeasts, fungi or bacteria used as a substitute for protein-rich foods, in human and animal feeds



Biomass production. The organism is grown under steady state conditions for periods of several weeks in a CSTR type fermenter. The culture medium consists of carbohydrate nutrient (hydrolyzed starch derived from potato, wheat or maize or molasses after removal of inorganic salts and gums) and other constituents such as ammonium phosphate, sulphates of potassium, magnesium, iron, zinc, manganese and copper, biotin, choline chloride, sulphuric acid (10%) and ammonia gas (fed directly to the fermenter via air stream). The blended culture medium is sterilized and continuously pumped into the fermenter together with metered amounts of air and ammonia gas. The medium is then inoculated and fermented, the biomass formed leaving the fermenter at a rate equal to the inflow of culture medium. The biomass is separated in a continuous cyclone separator into a slurry stream (allowing the gas stream to escape), which is passed on to an aseptic RNA reduction stage.

RNA reduction. The RNA reduction of the biomass increases the protein percentage of the final product. The reduction is achieved by using the endogenous Rnase of the cell to degrade its own RNA. A heat shock given to the cells initiates a series of complex responses which include the inactivation of protease and Rnase inhibitor, disruption of ribrosomes to release of RNA and degradation of RNA into nucleotides and nucleosides by Rnase. The cell loses its viability. The degradation products of RNA diffuse out through the cell wall. The RNA content is thereby reduced to about 1% from its initial value of about 10% on a dry weight basis. In addition, compounds with low molecular weight, such as amino acids and TCA cycle intermediates also diffuse out from the cell resulting in a total loss of about 35% of the initial cell dry weight. Because of this elimination of other intermediates the protein percentage in the final product is effectively higher compared to the protein content in the intact cells.

The RNA reduction is carried out aseptically in a stirred vessel provided with a steam jacket. Sterile air is supplied to the vessel head space to maintain a positive pressure. The optimum temperature for RNA reduction process is about 64°C. The pH of the culture medium usually remains unaltered at 6 during this process.

Recovery of biomass. The RNA reduced slurry is concentrated from about 1% to about 30% on dry weight basis by mechanical filter presses or centrifuges. The filter cake, called the mycoprotein, is similar to any moist perishable food and can be stored at 25°C in closed unsealed container for about 16 hours or at 4°C for about 48 hours. The product may be spray dried for storage at room temperature for months or even years. The mycoprotein is amenable for formulation into a variety of foods.

FOOD CHEMICALS:

Some of the important food chemicals are:

Amino acids

Amino acids. Amino acids are essential nutrients in foods, and are produced through various processes such as fermentation, enzymatic synthesis and bioconversion. Strains of *Corynebacterium, Arthrobacter, or Brevibacte-rium* lack the enzyme α-ketoglutarate dehydrogenase and hence glutamate accumulates in the cells due to transamination of α-ketoglutarate. Hence these organisms are used in the production of glutamic acid. Most of the amino acids are produced by enzymatic or bioconversion processes. Glutamic acid is used as a flavour enhancer (as monosodium glutamate, MSG) in food products. Alanine and glycine are used as flavours. Aspartic acid and phenylalanine find use in the production of the sweetener aspartame. Lysine, threonine and tryptophan are used as dietary supplements and cysteine as a reducing agent (antioxidant) in food industry.

Enzymes

Enzymes. The changes brought about in foods during microbial fermentation are primarily due to the various enzymes produced by the organisms. It is possible to isolate, purify and even immobilize the specific enzymes required for a desired function. Microbial enzymes produced by bacteria, yeasts and molds and their application in food industry is summarized in Table. 10.2

Lactic acid

Lactic acid. Lactic acid is produced industrially by homofermentative lactic acid bacteria. The bacteria used depends on the raw material to be fermented. *Lactobacillus delbrueckii* produces lactic acid from glucose,

Citric acid

Although citric acid is found in high concentrations in many citrus fruits, it is not economical to extract the acid from fruit for industrial use. Furthermore, the demand for citric acid far outweighs the supply of citrus fruit available.

The ability of the mold Aspergillus niger to produce citric acid as a byproduct of metabolism was discovered by American food chemist James Currie in 1917. The process of cultivating *A. niger* and allowing it to metabolize sucrose or glucose to yield citric acid proved efficient and inexpensive.

Vinegar

Vinegar production
•Acetic acid production by acetic acid bacteria
•Group of Gram negative bacteria that change ethanol to acetic acid under aerobic condition
•Acetobacter and Gluconobacter

Dextran and xantham

- Both are used as thickener and stabilizer.
- For the production of dextran the microbe Leucocnostoc mesenteroids is used
- For the production of xanthum the microbe Xanthomonas campestris is used

Beta carotene

Beta carotene. The microalgae *Dunaliella salina* is cultivated (similar to spirulina) as it is a rich source of beta carotene which is a natural antioxidant capable of neutralizing free radicals generated during oxidation reactions within the body.

UNIT- IV
FOOD BORNE DISEASES

The cause of food borne diseases is mainly the ingestion of infected food.
The general food related diseases such as gastrointestinal problems, improper diet, nutritional deficiencies and overheating.
Main cause of food borne diseases is due to
(i) food infection (Pathogenic organisms make their entry by consuming contaminated foods)
(ii) food intoxication (food poisoning is caused by consuming food contaminated with toxic substances)

FOOD INFECTIONS:

Broadly sub-classified into two types

i. Contaminated food does not ordinarily support the growth of the pathogenic organisms but merely carries them. (this include diphtheria, dysentery, typhoid, cholera, tuberculosis, infectious hepatitis)

ii. Food serves as a culture medium for the growth of pathogens to increase in number and cause infection upon consumption (salmonellosis, shigellosis, gastroenteritis)

Salmonellosis: Due to ingestion of viable cells of *Salmonella sp.*
Gram-negative, non-spore forming rod shaped bacteria, ferment glucose with gas formation and do not ferment lactose or sucrose. Survive a wide temperature range of 10-45 deg C with a pH 4-9. Holding the food at 66deg C for 12 min kills, source-unrefrigerated meat and poultry products.
After 12-36 hrs of infection, symptoms such as nausea, vomiting, abdominal pain, diarrhea, greenish foul smelling stools etc. Infections avoided by eliminating dead animals, cooking and heating the food for sufficient time and good refrigeration for storing.

***Clostridium Perfringens* Gastroenteritis:** Disease is due to entero-toxin released in the gut during sporulation of the cells. It is a gram positive anaerobic spore forming rod shaped bacteria, grows in temperature 20-50degC and pH 5-9. The spores show heat resistance. It occurs when raw foods and cooked meats allowed to cool slowly and held for some time before consumption. Cooking process kills vegetative cells and heat survived spores germinates when the food is not properly refrigerated. It develops abdominal pain, diarrhea and gas formation. Proper cooking, rapid cooling, reheating leftover foods are the prevention methods.

***Bacillus cereus* Gastroenteritis:** Gram positive aerobic spore forming rods- temp10-49 deg C, pH range 4.9-9.3. The contaminated food gives nausea abdominal cramps, vomiting and diarrhoea. Sources: mashed potatoes, meat loaf, puddings and custards. Prevention: proper cooking and reheating leftover foods upto 72degC.

***E.coli* Infection:** Enteropathogenic *E.coli* poisons foods cause human diarrhoeal diseases with vomiting, dehydration similar to cholera within 8-24hrs. Prevention: cooking foods thoroughly, rapid chilling of foods, using protected water.

Shigellosis: Bacillary dysentery caused by ingestion of moist infected foods such as milk, potato, shrimp, tuna turkey. Symptoms: abdominal cramps, chills, watery stools containing blood. Prevention: control of flies, purified water, good sanitary conditions etc.

Other Food Borne Infections:

Yersiniosis: (yersinia sp. caused by eating contaminated pork)

Vibrio: caused by eating raw marine foods, shell fish, crustacean

Hemolytic streptococcal infection: results in scarlet fever septic sore throat, cause by taking contaminated milk, ice creams, eggs and steamed lobster.

Viral Infections: infection by poliomyelitis and infectious hepatitis.

Parasitic infections:

Amebiasis: (Entamoeba histolytica) due to sewage contaminated water resulting in diarrhoea.

Trichinosis: (nematodes) due to raw or insufficiently cooked contaminated pork and port products, resulting in nausea, vomiting, diarrhoea, fever, swelling of eye lids.

Tapeworm: abdominal pain, digestive disorders due to improper cooked beef, fish or pork.

FOOD INTOXICATION:

Ingestion of contaminated food due to the presence of poisons or toxicants causes food poisoning. Classified into 4 types depending on the type of poisons 1. Bacterial 2. Fungal 3. Biological 4. Chemicals.

Bacterial Food Intoxication: refers to illness caused by the presence of bacterial toxin in the food.

This includes **Botulism** and **Staphylococcal intoxication**

BOTULISM: ingestion of food containing the neurotoxin produced by *Clostridium botulinum*. It is soil organism, spore former, gas forming anaerobic organism. Serologically there are 7 types of toxins produced. Type F is identified to cause human animals. The organisms grow and autolyze in the food to release toxin. Canned foods support the growth and toxin production. Optimal temperature for growth 26 to 35 deg C, moisture content more than 30% and pH above 4.5. Sodium chloride more than 8% inhibits growth and toxin production.

The toxin gets destroyed at 120 deg C for 4 minutes. It also gets destroyed by gamma radiation. This toxin, which is a protein of high potency, is absorbed in the small intestine and paralyzes the involuntary muscles of the body. The toxin paralyzes the involuntary

muscles and paralysis spread to heart and respiratory system and death results due to respiratory failure.

Foods inadequately heat processed canned foods such as vegetables, fruits, fish, and meat products. The preventive measure is to avoid consumption of canned foods which show indication of spoilage and any pressure in the container. The symptoms showed in 12 to 36 hours, which includes double vision and dryness of mouth and constriction of throat, swollen tongue and difficulty in swallowing and speaking. Administration of an antitoxin, particularly before the severe symptoms is the known treatment.

STAPHYLOCOCCAL GASTROENTERITIS: *S. aureus* produce an enterotoxin which causes gastroenteritis or inflammation of the lining of the intestine. The organisms are facultative but grow better in aerobic and induce coagulation of blood plasma. The strain tolerates sodium chloride to about 10-20%. They can grow even at meat curing solution and even on cured meat, even at sugars about 50% . Serologically there are 6 types of toxins, and type A is more poisonous. The toxin is stable up to 60-65 deg C for about 15 mins.

The enterotoxin is a protein with molecular weight about 30kda. Normal cooking of foods, or pasteurization at 72deg C or ultra high temperature heating at 143 deg C are not sufficient to render the toxin inactive. Ingestion of contaminated food leads to excess salivation, nausea, vomiting, abdominal cramps and diarrhea with blood and mucus and vomiting.

E.coli poisoning: The strain 157:H7 causes infection resuting bloody diarrhea and kidney failure.

MYCOTOXIN (Fungal toxins): They are fungal metabolits, potentially toxic to humans. The fungi include molds, yeasts, mushrooms. The species of the genera penicillium and aspergillus are known to produce mycotoxins such as aflatoxdins, patulin, ochratoxin, penicillic acid.

Aflatoxins: produced from Aspergillus flavus and A. parasiticus. Major aflatoxins have been designated as B1 and G1, shows the fluioresce blue and green respectively when exposed to UV radiation. B1 is more toxic one and others are carcinogenic to mammals and poultry. The fruit juices, cereals, peanuts and peanut meal and cottonseed meal

supports the growth of moulds. They grow and contaminates the foods during harvest itself and outbreaks food poisoning.

Patulin: produces by several molds such as *pencillium patulum*. They are first isolated as antibiotic to many bacterial species. They are toxic to seeds and seedling of heat peas, tomatoes resulting in pathological changes in organs of mice and rats and in higher concentration it is fatally toxic. It is a potential human carcinogen shows heat resistance even at 100 deg C.

Ochratoxin: Molds such as A. ostianus, P. cuclopium prodcuce ochratoxin which is toxic to pigs, rabbits and rats cause nephrotoxicity to kidneys.

CHEMICAL POISONING: usually characterized by appearance of the symptoms within short time after eating the poisoned food. Arsenic, antimony, cadmium, chlorinated hydrocarbons, copper, cyanide, fluoride, lead and nicotinic acid may enter foods from utensils, processing equipment, pesticides or insecticide spray residues on vegetables and futis.

FOOD SPOILAGE

Introduction:

Fitness of food is judged on the basis of individual consumer's acceptance, based on the visual characteristics and personal judgment of the consumers. Food acceptance criteria includes as desired stage of development or maturity – fruits and vegetables ripeness, poultry meat preferable from young birds. Freedom from microorganisms and endogenous food enzymes, over ripened fruits and vegetables.

Foods undergo deterioration to varying degrees i8n their organoleptic properties, nutritional values, safety and aesthetic appeal. They are subjected to physical, chemical and biological changes during spoilage.

Food leading to spoilage is a natural process and cannot be prevented but only delayed. From the moment food is harvested or slaughtered, it undergoes progressive deterioration in its quality and depending on the type of food, i.e may be slow as in the case of seeds and nuts, or may be rapid in milk and meat.

Based on the spoilage foods grouped as 3 categories.

1. **Nonperishable foods**: do not spoil normally with proper handling and storing. (sugar, salt, dry beans)

2. **Semi perishable foods**: unspoiled for relatively long period under proper handling or storing (Potatoes and apples)

3. **Perishable foods**: spoil readily, unless special preservative methods (dairy foods, meat, fish)

Cause of food spoilage:

1. Microbes, their growth and activity
2. Action of native enzymes
3. Insects, rodents and parasites
4. Chemical reactions of food constituents
5. Environmental factors (temp, moisture, air and light)
6. Time

Activity of microorganisms:

Spoilage organisms available commonly in soil, water, skin of cattle, fruits and vegetables, processing equipment and within the intestines and body cavities of animals. Raw foods contain a variety of bacteria, yeasts and moulds which increases their numbers.

Food spoilage involves association of microbes such as Symbiotic: mutually helpful, without hindering or aiding each other. Mutually antagonistic: antibiotic to each other, Synergetic: growing together to bring changes, which could never be produced when grown alone. (*pseudomonas sp on milk gives a light brown colour, with streptococcus sp gives brilliant blue colour*)

Microbes exhibit **Metabiotic effect**: one microbe grows and makes the condition favorable for the growth of second organism. (raw milk supports acid fermentation by *streptococcus lactis* and *coliform* bacteria until the acid formed inhibits their growth. Then acid tolerant lactobacilli grow producing more acid. Finally yeast and molds grow over the top and reduce the acidity facilitating the growth of proteolytic bacteria at a subsequent stage)

Moulds grow as fine threads of mat on the surface of the food. They also form spores that can be transported in air and infect other foods. They grow at room temperature and pH 4-6. At low temperature they grow slow but not inhibited. Psychrophillic molds grow even at refrigerated temperatures. Rhizopus grows on bread, Aspergillus and Penicillium grows to form soft rot in fruits. Upon growth, molds produce mycotoxins. (aflatoxin produced by aspergillus on cereals and peanuts)

Action of Native Enzymes:

Endogenous enzymes in plant and animal foods often intensified after harvest or slaughter due to lack of control mechanisms. Eg. Action of pepsin in digestion, but in dead animals digests organs, ripening of vegetables goes uncontrolled leads of spoilage. Enzymatic ripening and tenderization of animals are desirable but leads to deterioration due to rotting.

Insects, parasites and rodents:

Insects destroy cereals grains, fruits and vegetables by contamination and facilitate microbial attack. Fumigation of stored areas with ethylene oxide and methyl bromide controls the insects. Parasites enter in human body through poultry and meat which have been improperly cooked. Rodents also consume considerable quantity of food and contaminate through droppings, urine and pathogenic bacteria.

Chemical reactions:

Food quality deteriorates due to chemical reactions. The unsaturated fatty acid components undergo oxidation due to exposure to atmospheric air giving rise to oxidative

rancidity in fat rich foods causing odour and changes in texture of food. Light induced reduction of vitamins is seen due to oxidation.

Environmental factors (temp, moisture and humidity, air and light**):**

Excessive heat denatures proteins, breaks emulsions, vitamins and enhances the rates of chemical and enzymatic reactions and dries out food by removing moisture. Excessive cold also spoils liquid foods. Moisture content and humidity in atmosphere decrease quality deterioration. Even moisture proof pickings, moisture gets trapped by respiration of fruits and vegetables. Air induces the oxidation. Light destroys riboflavin, vitamin A, C and promotes oxidation

Time:

The quality of foods remains maximum for some time soon after harvest and slaughter (freshness) thereafter deteriorates as time progresses. It is better to keep vegetable unplucked and also keep alive the fish and poultry till the time of need. Soon after the harvest, the food should be cleaned an cooled rapidly.

SPOILAGE OF VARIOUS FOODS AND FOOD PRODUCTS

Vegetables and Fruits: Bacterial soft rot of fruits and vegetables is caused by *Erwinia carotovora* and *pseudomonas marginalis*. They break down pectins, giving rise to a soft, mushy consistency, bad odour and water soaked appearance. (eg. Onion, bean, carrot, tomato, cucumber and radish).

Sour rot is affected to bean, carrot lettuce, cabbage, cauliflower etc.by Geotrichum candidum

Black rot of cabbage and cauliflower (Xanthomonas campestris), ring rot of potatoes (Geotrichum candidum). Slimy brown rot of vegetables (Phytopora sp.)

Cereals and Cereal Products: Low water content of wheat, rice, corn protects them from microbial attack if stored properly. Under favorable conditions of water activity, bacteria of the genus Bacillus and molds grow. Many of the aerobic spore formers produce amylase and hence are capable of utilizing starch in flours.

Bakery Products: The moisture content of commercially made bread is insufficient for the growth of most organisms except molds. Rhizopus stolonifer is the common bread mold causing the spoilage (ropiness, hardening in bread) when stored in high humidity. Cakes with high sugar content does not allow bacterial spoilage. The baking temperature is sufficient to destry many of these organism but toppings, icings, fruits are the source of molds.

Dairy Products: Milk, cream, butter and Cheese are susceptible to microbial spoilage. Milk is growth medium for many organism such as mycobacterium,

propionibacterium, micrococcus and other coliforms. Pasteurization eliminates many of the species. Heat resistant streptococci cause spoilage even after heating and produce lactic acid by fermenting lactose and decrease pH to about 4.5 leading to curdling of milk.

Butter undergoes microbial spoilage as surface taint or **putridity** and **rancidity**. Putridity is caused by *Pseudomonas putrefaciens* on the surface of the butter producing organic acids particularly isovaleric acid, causing bad odours. Rancidity (*p.fragi*) occurs due to hydrolysis of butter fat resulting in the formation of free fatty acids. Pseudomonas mephitica cause black discoloration and skunk-like odour.

Cheese gets spoiled to a condition called slimy curd. Gassiness of cheese is caused due to the formation of carbon dioxide formed by lactose utilizing organisms of Clostridium sp.

Meat, Poultry and seafoods: Fresh meat is spoiled by micro flora coming from animal lymph nodes, intestinal tract, hide and handling equipment. Mucor and Rhizophus are the fungal spoilage on meat giving green and black patches. Fresh liver, pork and lamb have higher content of carbohydrates and near neutral pH value undergo fermentative spoilage by lactic acid bacteria. Poultry mostly spoiled only on surface as the inner portions of poultry tissue are generally sterile which leads to off odour and slimy appearance. Fish (fresh water and salt water) contain high level of proteins and low fat content and practically no carbohydrates. Fresh and iced fish spoiled by bacteria and salted and dried fish spoiled by fungi. Intestinal bacteria and native proteolytic enzymes also contribute to the spoilage and develop sliminess and off-odors. Hence fish need to be eviscerated immediately. Shell fish such as shrimp, lobster, crab undergo bacterial spoilage and fermentative spoilage due to its high carbohydrate content.

Eggs: Hen's egg is well protected by 3 structures (an outer waxy shell membrane, the shell and inner shell membrane). Egg white is not susceptible to microbial spoilage due to its pH 9.3 and its antimicrobial property. It contains lysozyme which is effective against microbs and avidin which forms a complex with biotin rendering the vitamin unavailable to microbes and conalbumins forms complex with iron preventing its use to microbes. Its yolk is medium for the microbial growth because its neutral pH and nitrogen content. Bacterial spoilage leads to hydrogen sulphide and watery and discolouration of yolk.

Fresh eggs are sterile and prone to microbes (pseudomonas, acinetobacter, proteus, micrococcus, salmonella, enterobacter) when stored in humid condition and facilitates microbial penetration into the shell.

Confectionery items: under improper storage and high humidity, torula and osmophilic saccharomyces can cause inversion of sugar (converts sucrose to dextran, a gummy and slimy polymer of glucose)

Alcoholic beverages: Beer and wine infections are caused by bacteria and yeasts. Four different spoilage such as ropiness, sarcinase sickness, sourness and turbidity. Ropiness: caused by Acetobacter and Lactobacillus leads to beer becoming viscous and oily. Sarcinase sickness: caused by P. cerevisiae leading diacetyl compound production and giving rise to honey like odour to the beer. Sourness: caused ny Acetobacter due to the acetic acid formed by the oxidation of ethanol. Turbidity and off odour by Zymononas anaerobia. Yeast does Malo-lactic fermentation of grapes and its malic and tartaric acid content reduced to form off quality wines.

UNIT V

STERILIZATION

The time and temperature required for the sterilization of foods are influenced by several factors, including the type of microorganisms found on the food, the size of the container, the acidity or pH of the food, and the method of heating.

The thermal processes of canning are generally designed to destroy the spores of the bacterium C. botulinum. This microorganism can easily grow under anaerobic conditions, producing the deadly toxin that causes botulism. Sterilization requires heating to temperatures greater than 100° C (212° F). However, C. botulinum is not viable in acidic foods that have a pH less than 4.6. These foods can be adequately processed by immersion in water at temperatures just below 100° C.

The sterilization of low-acid foods (pH greater than 4.6) is generally carried out in steam vessels called retorts at temperatures ranging from 116° to 129° C (240° to 265° F). The retorts are controlled by automatic devices, and detailed records are kept of the time and temperature treatments for each lot of processed cans. At the end of the heating cycle, the cans are cooled under water sprays or in water baths to approximately 38° C (100° F) and dried to prevent any surface rusting. The cans are then labeled, placed in fibre board cases either by hand or machine, and stored in cool, dry warehouses.

QUALITY OF CANNED FOODS

The sterilization process is designed to provide the required heat treatment to the slowest heating location inside the can, called the cold spot. The areas of food farthest from the cold spot get a more severe heat treatment that may result in over processing and impairment of the overall quality of the product. Flat, laminated pouches can reduce the heat damage caused by over processing.

A significant loss of nutrients, especially heat-labile vitamins, may occur during the canning process. In general, canning has no major effect on the carbohydrate, protein, or fat content of foods. Vitamins A and D and beta-carotene are resistant to the effects of heat. However, vitamin B1 is sensitive to thermal treatment and the pH of the food. Although the anaerobic conditions of canned foods have a protective effect on the stability of vitamin C, it is destroyed during long heat treatments.

The ends of processed cans are slightly concave because of the internal vacuum created during sealing. Any bulging of the ends of a can may indicate deterioration in quality due to mechanical, chemical, or physical factors. This bulging may lead to swelling and possible explosion of the can.

Pasteurization

Pasteurization is the application of heat to a food product in order to destroy pathogenic (disease-producing) microorganisms, to inactivate spoilage-causing enzymes, and to reduce or destroy spoilage microorganisms. The relatively mild heat treatment used in the pasteurization process causes minimal changes in the sensory and nutritional characteristics of foods compared to the severe heat treatments used in the sterilization process.

The temperature and time requirements of the pasteurization process are influenced by the pH of the food. When the pH is below 4.5, spoilage microorganisms and enzymes are the main targets of pasteurization. For example, the pasteurization process for fruit juices is aimed at inactivating certain enzymes such as pectinesterase and polygalacturonase. The typical processing conditions for the pasteurization of fruit juices include heating to 77° C (171° F) and holding for 1 minute, followed by rapid cooling to 7° C (45° F). In addition to inactivating enzymes, these conditions destroy any yeasts or molds that may lead to spoilage. Equivalent conditions capable of reducing spoilage microorganisms involve heating to 65° C (149° F) and holding for 30 minutes or heating to 88° C (190° F) and holding for 15 seconds.

When the pH of a food is greater than 4.5, the heat treatment must be severe enough to destroy pathogenic bacteria. In the pasteurization of milk, the time and temperature conditions target the pathogenic bacteria Mycobacterium tuberculosis, Coxiellaburnetti, and Brucellaabortus. The typical heat treatment used for pasteurizing milk is 72° C (162° F) for 15 seconds, followed by rapid cooling to 7° C. Other equivalent heat treatments include heating to 63° C (145° F) for 30 minutes, 90° C (194° F) for 0.5 second, and 94° C (201° F) for 0.1 second. The high-temperature–short-time (HTST) treatments cause less damage to the nutrient composition and sensory characteristics of foods and therefore are preferred over the low-temperature–long-time (LTLT) treatments.

Since the heat treatment of pasteurization is not severe enough to render a product sterile, additional methods such as refrigeration, fermentation, or the addition of chemicals are often used to control microbial growth and to extend the shelf life of a product. For example, the pasteurization of milk does not kill thermoduric bacteria (those resistant to heat), such as Lactobacillus and Streptococcus, or thermophilic bacteria (those that grow at high temperatures), such as Bacillus and Clostridium. Therefore, pasteurized milk must be kept under refrigerated conditions.

Liquid foods such as milk, fruit juices, beers, wines, and liquid eggs are pasteurized using plate-type heat exchangers. Wine and fruit juices are normally deaerated prior to pasteurization in order to remove oxygen and minimize oxidative deterioration of the products. Plate-type heat exchangers consist of a large number of thin, vertical steel plates that are clamped together in a frame. The plates are separated by small gaskets that allow the liquid to flow between each successive plate. The liquid product and heating medium (e.g., hot water) are pumped through alternate channels, and the gaskets ensure that the liquid product and heating or cooling mediums are kept separate. Plate-type heat exchangers are effective in rapid heating and cooling applications. After the pasteurization process is completed, the product is packaged under aseptic conditions to prevent recontamination of the product.

Aseptic processing

The aseptic process involves placing a sterilized product into a sterilized package that is then sealed under sterile conditions. It began in 1914 with the development of sterile filters for use in the wine industry. However, because of unreliable machinery, it remained commercially unsuccessful until 1948 when William McKinley Martin helped develop the Martin system, which later became known as the Dole Aseptic Canning System. This system involved the sterilization of liquid foods by rapidly heating them in tubular heat exchangers, followed by holding and cooling steps. The cans and lids were sterilized with superheated steam, and the sterilized containers were filled with the sterile liquid food. The lids were then sealed in an atmosphere of superheated steam. By the 1980shydrogen peroxide was being used throughout Europe and the United States for the sterilization of polyethylene surfaces.

COMMERCIAL STERILIZATION

In aseptic processing the thermal process is based on achieving commercial sterility—i.e., no more than 1 nonsterile package for every 10,000 processed packages. The aseptic process uses the high-temperature–short-time (HTST) method in which foods are heated at a high temperature for a short period of time. The time and temperature conditions depend on several factors, such as size, shape, and type of food. The HTST method results in a higher retention of quality characteristics, such as vitamins, odour, flavour, and texture, while achieving the same level of sterility as the traditional canning process in which food is heated at a lower temperature for a longer period of time.

The heating and cooling of liquid foods can be performed using metal plate heat exchangers. These heat exchangers have large surface areas that result in improved heating and cooling rates. Other types of heat exchangers involve surrounding the food with steam or directly injecting steam into the food. Products sterilized with steam are then pumped into a vacuum chamber, where they are cooled rapidly.

Liquid foods that contain large solid particles are heated in scraped-surface heat exchangers. These heat exchangers use blades to continuously scrape the inside surface of the heating chamber. The scraping action protects highly viscous foods from being burned on the heating surface.

An alternate method for heating foods, called ohmic heating, passes a low-frequency electric current of 50 to 60 hertz directly through the food. A liquid food containing solids, such as diced fruit, is pumped through a pipe surrounded by electrodes. The product is heated as long as the electrical conductivity of the food is uniform throughout the entire volume. This uniform rate of heating prevents the over processing of any individual region of the food. Ohmic heating yields a food product of higher quality than those processed using conventional systems.

PACKAGING ASEPTICALLY PROCESSED PRODUCTS

The packaging containers used in aseptic processing are sterilized separately before they are used. The packaging machinery is sterilized using steam, sterile gases, or hydrogen peroxide. The sterilization process is generally monitored by culturing a test organism.

For example, the remaining presence of the highly heat-resistant bacterium Bacillus subtilisglobigii can be used as a marker to measure the completeness of sterilization.

Packages must be sealed under sterile conditions, usually using high-temperature sealing plates. Foods that are aseptically processed do not require refrigeration for storage.

Blanching

Blanching is a thermal process used mostly for vegetable tissues prior to freezing, drying, or canning. Before canning, blanching serves several purposes, including cleaning of the product, reducing the microbial load, removing any entrapped gases, and wilting the tissues of leafy vegetables so that they can be easily put into the containers. Blanching also inactivates enzymes that cause deterioration of foods during frozen storage.

Blanching is carried out at temperatures close to 100° C (212° F) for two to five minutes in either a water bath or a steam chamber. Because steam blanchers use a minimal amount of water, extra care must be taken to ensure that the product is uniformly exposed to the steam. Steam blanching leafy vegetables is especially difficult because they tend to clump together. The effectiveness of the blanching treatment is usually determined by measuring the residual activity of an enzyme calledperoxidase.

Controlling water activity

Foods containing high concentrations of water are generally more susceptible to deterioration by microbial contamination and enzymatic activity. The water content of foods can be controlled by removing water through dehydration or by adding solutes to the food. In both cases the concentration of solutes in the food increases and the concentration of water decreases.

Dehydration

Dehydration, or drying, of foods has long been practiced commercially in the production of spaghetti and other starch products. As a result of advances made during World War II, the technique has been applied to a growing list of food products, including fruits, vegetables, skim milk, potatoes, soup mixes, and meats.

Pathogenic (toxin-producing) bacteria occasionally withstand the unfavourable environment of dried foods, causing food poisoning when the product is rehydrated and eaten. Control of bacterial contaminants in dried foods requires high-quality raw

materials having low contamination, adequate sanitation in the processing plant, pasteurization before drying, and storage conditions that protect from infection by dust, insects, and rodents or other animals.

Foodstuffs may be dried in air, superheated steam, vacuum, or inert gas or by direct application of heat. Air is the most generally used drying medium, because it is plentiful and convenient and permits gradual drying, allowing sufficient control to avoid overheating that might result in scorching and discoloration. Air may be used both to transport heat to the food being dried and to carry away liberated moisture vapour. The use of other gases requires special moisture recovery systems.

Loss of moisture content produced by drying results in increased concentration of nutrients in the remaining food mass. The proteins, fats, and carbohydrates in dried foods are present in larger amounts per unit weight than in their fresh counterparts, and the nutrient value of most reconstituted or rehydrated foods is comparable to that of fresh items. The biological value of driedprotein is dependent, however, on the method of drying. Prolonged exposure to high temperatures can render the protein less useful in the diet. Low-temperature treatment, on the other hand, may increase the digestibility of protein. Some vitamins are sensitive to the dehydration process. For example, in dried meats significant amounts of vitamin C and the B vitamins—riboflavin, thiamine, and niacin—are lost during dehydration.

Dried eggs, meat, milk, and vegetables are ordinarily packaged in tin or aluminum containers. Fibreboard or other types of material may be employed but are less satisfactory than metal, which offers protection against insects and moisture loss or gain and which permits packaging with an inert gas.

In-package desiccants (drying agents) improve storage stability of dehydrated white potatoes, sweet potatoes, cabbage, carrots, beets, and onions and give substantial protection against browning. Retention of ascorbic acid (vitamin C) is markedly improved by packaging at temperatures up to 49° C (120° F); the packaging gas may be either nitrogen or air.

A related technique, freeze-drying, employs high vacuum conditions, permitting establishment of specific temperature and pressure conditions. The raw food is frozen,

and the low pressure conditions cause the ice in the food to sublimate directly into vapour (i.e., it does not transit through the liquid state). Adequate control of processing conditions contributes to satisfactory rehydration, with substantial retention of nutrient, colour, flavour, and texture characteristics.

Concentration of moist foods

Foods with substantial acidity, when concentrated to 65 percent or more soluble solids, may be preserved by mild heat treatments. High acid content is not a requirement for preserving foods concentrated to over 70 percent solids.

Fruit jelly and preserve manufacture, an important fruit by-product industry, is based on the high-solids–high-acid principle, with its moderate heat-treatment requirements. Fruits that possess excellent qualities but are visually unattractive may be preserved and utilized in the form of concentrates, which have a pleasing taste and substantial nutritive value.

Jellies and other fruit preserves are prepared from fruit by adding sugar and concentrating by evaporation to a point where microbial spoilage cannot occur. The prepared product can be stored without hermetic sealing, although such protection is useful to control mold growth, moisture loss, and oxidation. In modern practice, vacuum sealing has replaced the use of a paraffin cover.

The jelly-forming characteristics of fruits and their extracts are due to pectin, a substance present in varying amounts in all fruits. The essential ingredients in a fruit gel are pectin, acid, sugar, and water. Flavouring and colouring agents may be added, and additional pectin and acid may be added to overcome any deficiencies in the fruit itself.

Candied and glacéed fruits are made by slow impregnation of the fruit with syrup until the concentration of sugar in the tissue is sufficiently high to prevent growth of spoilage microorganisms. The candying process is conducted by treating fruits with syrups of progressively increasing sugar concentrations, so that the fruit does not soften into jam or become tough and leathery. After sugar impregnation the fruit is washed and dried. The resulting candied fruit may be packaged and marketed in this condition or may be dipped into syrup, becoming coated with a thin glazing of sugar (glacéed) and again dried.

Fermentation and pickling

Although microorganisms are usually thought of as causing spoilage, they are capable under certain conditions of producing desirable effects, including oxidative and alcoholic

fermentation. The microorganisms that grow in a food product, and the changes they produce, are determined by acidity, available carbohydrates, oxygen, and temperature. An important food preservation method combines salting to control microorganisms selectively and fermentation to stabilize the treated tissues.

Pickled fruits and vegetables

Fresh fruits and vegetables soften after 24 hours in a watery solution and begin a slow, mixed fermentation-putrefaction. The addition of salt suppresses undesirable microbial activity, creating a favourable environment for the desired fermentation. Most green vegetables and fruit may be preserved by pickling.

When the pickling process is applied to a cucumber, its fermentable carbohydrate reserve is turned into acid, its colour changes from bright green to olive or yellow-green, and its tissue becomes translucent. The salt concentration is maintained at 8 to 10 percent during the first week and is increased 1 percent a week thereafter until the solution reaches 16 percent. Under properly controlled conditions the salted, fermented cucumber, called salt stock, may be held for several years.

Salt stock is not a consumer commodity. It must be freshened and prepared into consumer items. In cucumbers this is accomplished by leaching the salt from the cured cucumber with warm water (43°–54° C [110°–130° F]) for 10 to 14 hours. This process is repeated at least twice, and, in the final wash, alum may be added to firm the tissue and turmeric to improve the colour.

Pickled meat

Meat may be preserved by dry curing or with a pickling solution. The ingredients used in curing and pickling are sodium nitrate, sodium nitrite, sodium chloride, sugar, and citric acid or vinegar.

Various methods are used: the meat may be mixed with dry ingredients; it may be soaked in pickling solution; pickling solution may be pumped or injected into the flesh; or a combination of these methods may be used.

Curing may be combined with smoking. Smoke acts as a dehydrating agent and coats the meat surfaces with various chemicals, including small amounts of formaldehyde.

Deterioration of fermented and pickled products

Fermented foods and pickled products require protection against molds, which metabolize the acid developed and allow the advance of other microorganisms. Fermented and pickled food products placed in cool storage can be expected to remain stable for several months. Longer storage periods demand more complete protection, such as canning.

Nutrient retention in fermented and pickled products is about equal to retention for products preserved by other methods. Carbohydrates usually undergo conversion to acid or to alcohol, but these are also of nutritive value. In some instances, nutrient levels are increased because of the presence of yeasts.

Chemical preservation

Chemical food preservatives are substances which, under certain conditions, either delay the growth of microorganisms without necessarily destroying them or prevent deterioration of quality during manufacture and distribution. The former group includes some natural food constituents which, when added to foods, retard or prevent the growth of microorganisms. Sugar is used partly for this purpose in making jams, jellies, and marmalades and in candying fruit. The use of vinegarand salt in pickling and of alcohol in brandying also falls in this category. Some chemicals foreign to foods are added to prevent the growth of microorganisms. The latter group includes some natural food constituents such as ascorbic acid (vitamin C), which is added to frozen peaches to prevent browning, and a long list of chemical compounds foreign to foods and classified as antioxidants, bleaching agents, acidulants, neutralizers, stabilizers, firming agents, and humectants.

Organic chemical preservatives

Sodium benzoate and other benzoates are among the principal chemical preservatives. The use of benzoates in certain products in prescribed quantity (usually not exceeding 0.1 percent) is permitted in most countries, some of which require a declaration of its use on the label of the food container. Since free benzoic acid actually is the active agent, benzoates must be used in an acid medium in order to be effective. The ability of

cranberries to resist rapid deterioration is attributed to their high benzoic acid content. Benzoic acid is more effective against yeasts than against molds and bacteria.

Other organic compounds used as preservatives include vanillic acid esters, monochloroacetic acid, propionates, sorbic acid, dehydroacetic acid, and glycols.

Inorganic chemical preservatives

Sulfur dioxide and sulfites are perhaps the most important inorganic chemical preservatives. Sulfites are more effective against molds than against yeasts and are widely used in the preservation of fruits and vegetables. Sulfur compounds are extensively used in wine making and, as in most other instances when this preservative is used, much care has to be exercised to keep the concentrations low in order to avoid undesirable effects on flavour.

Oxidizing agents such as nitrates and nitrites are commonly used in the curing of meats.

Low-temperature preservation

Storage at low temperatures prolongs the shelf life of many foods. In general, low temperatures reduce the growth rates of microorganisms and slow many of the physical and chemical reactions that occur in foods.

Refrigeration

The life of many foods may be increased by storage at temperatures below 4° C (40° F). Commonly refrigerated foods include fresh fruits and vegetables, eggs, dairy products, and meats. Some foods, such as tropical fruits (e.g., bananas), are damaged if exposed to low temperatures. Also, refrigeration cannot improve the quality of decayed food; it can only retard deterioration. One problem of modern mechanical refrigeration—that of dehydration of foods due to moisture condensation—has been overcome through humidity control mechanisms within the storage chamber and by appropriate packaging techniques.

Freezing

Freezing and frozen storage provide an excellent means of preserving the nutritional quality of foods. At subfreezing temperatures the nutrient loss is extremely slow for the typical storage period used in commercial trade.

HISTORY

Early freezing methods were based on the principle that mixing salt with ice results in temperatures well below 0° C (32° F). By the end of the 19th century, this method was being used commercially in the United States to freeze fish and poultry. By the 1920s Clarence Birdseye had developed two processes for freezing fish based on his quick freezing theory. His first patent, describing a method for preserving piscatorial products, involved placing food between two metal plates that were chilled by a calcium chloride solution to approximately −40° C (−40° F). The second process utilized two hollow metal plates that were cooled to −25° C (−13° F) by vaporization of ammonia. This freezing apparatus was the forerunner of the multiple plate freezer that is widely used in the modern food industry.

THE FREEZING PROCESS

The freezing of food involves lowering its temperature below 0° C, resulting in the gradual conversion of water, present in the food, into ice. Freezing is a crystallization process that begins with a nucleus or a seed derived from either a nonaqueous particle or a cluster of water molecules (formed when the temperature is reduced below 0° C). This seed must be of a certain size to provide an adequate site for the crystal to begin to grow. If physical conditions are conducive to the presence of numerous seeds for crystallization, then a large number of small ice crystals will form. However, if only a few seeds are initially available, then a few ice crystals will form and each will grow to a large size. The size and the number of ice crystals influence the final quality of many frozen foods; for example, the smooth texture of ice cream indicates the presence of a large number of small ice crystals.

In pure water, the freezing process is initiated by lowering the temperature to slightly below 0° C, called supercooling. As ice crystals begin to grow, the temperature returns to the freezing point. During the conversion of liquid water to ice, the temperature of the system does not change. The heat removed during this step is called the latent heat of fusion (equivalent to 333 joules per gram of water). Once all the water is converted to ice, any additional removal of heat will result in a decrease in the temperature below 0° C.

The freezing of foods exhibits a number of important differences from the freezing of pure water. Foods do not freeze at 0° C. Instead, owing to the presence of different soluble particulates (solutes) in the water present in foods, most foods begin to freeze at a temperature between 0° and −5° X (32° and 23° F). In addition, the removal of latent heat in foods during freezing does not occur at a fixed temperature. As the water present in the food freezes into ice, the remaining water becomes more concentrated with solutes. As a result, the freezing point is further depressed. Therefore, foods have a zone of maximum ice crystal formation that typically extends from −1° to −4° C (30° to 25° F). Damage to food quality during freezing can be minimized if the temperature of the product is brought below this temperature range as quickly as possible.

INDUSTRIAL FREEZERS

The rate at which heat is removed from a food during freezing depends on how fast heat can travel within the food and how efficiently it can be liberated from the surface of the food into the surrounding atmosphere. Industrial freezers remove heat from the surface of a food as rapidly as possible. There are several types of industrial freezers, including air-blast tunnel freezers, belt freezers, fluidized-bed freezers, plate freezers, and cryogenic freezers.

In air-blast tunnel freezers and belt freezers, precooled air at approximately −40° C is blown over the food products. Packaged foods, such as fruits, vegetables, bakery goods, poultry, meats, and prepared meals, are usually frozen in air-blast tunnels. The packages are placed onto dollies or hand trucks and then rolled into the freezer tunnels. In a belt freezer, food is placed on a conveyor belt that moves through a freezing zone. Bakery goods, chicken parts, and meat patties are frozen using a belt freezer.

Fluidized-bed freezers are used to freeze particulate foods such as peas, cut corn, diced carrots, and strawberries. The foods are placed on a mesh conveyor belt and moved through a freezing zone in which cold air is directed upward through the mesh belt and the food particulates begin to tumble and float. This tumbling exposes all sides of the food to the cold air and minimizes the resistance to heat transfer at the surface of the food.

Plate freezers are used to freeze flat products, such as pastries, fish fillets, and beef patties, as well as irregular-shaped vegetables that are packaged in brick-shaped containers, such as asparagus, cauliflower, spinach, and broccoli. The food is firmly pressed between metal plates that are cooled to subfreezing temperatures by internally circulating refrigerants.

Cryogenic freezing is used to freeze food at an extremely fast rate. The food is moved through a spray of liquid nitrogen or directly immersed in liquid nitrogen. The liquid nitrogen boils around the food at a temperature of −196° C (−321° F) and extracts a large amount of heat.

QUALITY OF FROZEN FOODS

Improper freezing or storage of foods may result in detrimental quality changes. When foods with high amounts of water are frozen slowly, they may experience a loss of fluid, called drip, upon thawing. This fluid loss causes dehydration and nutrient loss in frozen food products.

During frozen storage, the ice crystals present in foods may enlarge in size, producing undesirable changes in texture. This phenomenon is commonly observed when the storage temperature is allowed to fluctuate. For example, ice cream stored in an automatic defrosting domestic freezer becomes sandy in texture because the ice crystals increase in size as the temperature of the system fluctuates.

Improperly packaged frozen foods lose small amounts of moisture during storage, resulting in surface dehydration (commonly called freezer burn). Frozen meats with freezer burn have the appearance of brown paper and quickly become rancid. Freezer burn can be minimized by the use of tightly wrapped packages and the elimination of fluctuating temperatures during storage.

Food irradiation

Food irradiation is the process of treating food with a specific dosage of ionizing radiation. This treatment slows or halts spoilage by retarding enzymic action or destroying microorganisms and it can also inactivate foodborne pathogenic organisms (reducing the risk of food borne illness). Further applications include sprout inhibition,

delay of ripening, increase of juice yield, and improvement of re-hydration. Irradiation is also used to prevent the spread of invasive insect species that could be associated with fresh produce (e.g. fruit fly pests).

Ionizing radiation affects cells and microorganisms by damaging their DNA beyond its ability to repair, breaking down cell membranes and interrupting enzymic pathways. Organisms can no longer successfully continue the process of cell division. The major effect of irradiation is to generate short-lived and transient radicals (e.g. the hydroxy radical, the hydrogen atom and solvated electrons) that in turn damage DNA and intercellular structures. The target organism ceases all processes related to maturation or reproduction. At high enough doses the target organism does not survive. Irradiated food does not become radioactive, as the particles that transmit radiation are not themselves radioactive. Still there is some controversy in the application of irradiation possibly due to it being confused with radioactive contamination and an imputed association with the nuclear industry, and also perhaps because people think that chemical changes will be different than the chemical changes due to heating food (as ionizing radiation produces a higher energy transfer per collision than conventional radiant heat). However, research has found that most, if not all, of the substances formed in irradiated food are also found in food that has been subjected to other processing treatments and are not exclusively formed by irradiation. Furthermore, the quantities in which they occur in irradiated food are not significantly higher than those being formed in heat treatments.

Irradiation is a more general term of the exposure of materials to radiation to achieve a technical goal (in this context "ionizing radiation" is implied). As such it is also used on non-food items, such as medical devices, plastics, tubes for gas pipelines, hoses for floor heating, shrink-foils for food packaging, automobile parts, wires and cables (isolation), tires, and even gemstones.

Processing of food by ionizing radiationBy irradiating food, depending on the dose, some or all of the microorganisms, bacteria, viruses, or insects present are destroyed or rendered incapable of reproduction. Hence pathogenic organisms can be inactivated, reducing the risk of food borne illness, and spoilage organisms can also be severely affected, thereby prolonging the shelf-life of the food. Some foods, e.g., herbs and spices,

are irradiated at sufficient doses (five kilograys or more) to reduce the microbial counts by several orders of magnitude; such ingredients do not carry over spoilage or pathogen microorganisms into the final product. It has also been shown that irradiation can delay the ripening of fruits or the sprouting of vegetables.

Food irradiation is sometimes referred to as "cold pasteurization" or "electronic pasteurization"because ionizing the food does not heat the food to high temperatures during the process, as in heat-pasteurization (at a typical dose of 10 kGy, food that is physically equivalent to water would warm by about 2.5 °C). The treatment of solid food by ionizing radiation can provide an effect similar to heat pasteurization of liquids, such as milk. The use of the term "cold pasteurization" to describe irradiated foods is controversial, because pasteurization and irradiation are fundamentally different processes, although the intended end results can in some cases be similar.

Radappertization is a form of food irradiation which applies a dose of ionizing radiation sufficient to reduce the number and activity of viable microorganisms to such an extent that very few, if any, are detectable in the treated food by any recognized method (viruses being excepted). No microbial spoilage or toxicity should become detectable in a food so treated, regardless of the conditions under which it is stored, provided the packaging remains undamaged. The required dose is usually in the range of 25-45 kGy.

Radicidation is a specific case of food irradiation where the dose of ionizing radiation applied to the food is sufficient to reduce the number of viable specific non-spore-forming pathogenic bacteria to such a level that none are detectable when the treated food is examined by any recognized method. The required dose is in the range of 2 - 8 kGy. The term may also be applied to the destruction of parasites such as tapeworm and trichina in meat, in which case the required dose is in the range of 0.1 - 1 kGy.

Radurization is the application to food irradiation to foods using a dose of ionizing radiation sufficient to enhance its keeping quality by causing a substantial decrease in numbers of viable specific spoilage microorganisms. The required dose is in the range of 0.4 - 10 kGy.

Efficiency illustration of the different radiation technologies (electron beam, X-ray, gamma rays)[edit] Electron irradiationElectron irradiation uses electrons accelerated in

an electric field to a velocity close to the speed of light. Electrons are particulate radiation and, as any other particulate radiation, have a limited range in matter. For this reason, electrons do not penetrate the product beyond a few centimeters, depending on product density. However, electron facilties (like gamma and x-ray facilities) rely on substantial concrete shields to protect workers and the environment from radiation exposure.

Gamma radiation is radiation of photons in the gamma part of the electromagnetic spectrum. The radiation is obtained through the use of radioisotopes, generally cobalt-60 but caesium-137 can also be used as a source. Cobalt-60 is bred from cobalt-59 using neutron irradiation in specifically designed nuclear reactors. Caesium-137 is recovered during the processing of spent nuclear fuel. Because this technology – except for military applications – is not commercially available, insufficient quantities of it are available on the global isotope markets for use in large scale, commercial irradiators. Presently, caesium-137 is used only in small hospital units to treat blood before transfusion to prevent Graft-versus-host disease.

Food irradiation using cobalt-60 is the preferred method by most processors, because the deeper penetration enables administering treatment to entire industrial pallets or totes, reducing the need for material handling.A pallet or tote is typically exposed for several minutes to hours depending on dose. Radioactive material must be monitored and carefully stored to shield workers and the environment from its gamma rays. During operation this is achieved by substantial concrete shields. With most designs the radioisotope can be lowered into a water-filled source storage pool to allow maintenance personnel to enter the radiation shield. In this mode the water in the pool absorbs the radiation. Other uncommonly used designs feature dry storage by providing movable shields that reduce radiation levels in areas of the irradiation chamber.

Similar to gamma radiation, X-rays are photon radiation of a wide energy spectrum and an alternative to isotope based irradiation systems. X-rays are generated by colliding accelerated electrons with a dense material (target) such as tantalum or tungsten in a process known as bremsstrahlung-conversion. X-ray irradiators are scalable and have deep penetration comparable to Co-60, with the added benefit that the electronic source stops radiating when switched off. They also permit dose uniformity, but these systems

generally have low energetic efficiency during the conversion of electron energy to photon radiation requiring much more electrical energy than other systems. Like most other types of facilities, X-ray systems rely on concrete shields to protect the environment and workers from radiation.

The safety of irradiation facilities is regulated by the United Nations International Atomic Energy Agency and monitored by the different national Nuclear Regulatory Commissions. The incidents that have occurred in the past are documented by the agency and thoroughly analyzed to determine root cause and improvement potential. Such improvements are then mandated to retrofit existing facilities and future design.

The cost of food irradiation is influenced by dose requirements, the food's tolerance of radiation, handling conditions, i.e., packaging and stacking requirements, construction costs, financing arrangements, and other variables particular to the situation. In the case of large research or contract irradiation facilities, major capital costs include a radiation source (cobalt-60), hardware (irradiator, totes and conveyors, control systems, and other auxiliary equipment), land (1 to 1.5 acres), radiation shield, and warehouse. Operating costs include salaries (for fixed and variable labor), utilities, maintenance, taxes/insurance, cobalt-60 replenishment, general utilities, and miscellaneous operating costs.

Opponents of food irradiation sometimes state that large-scale irradiation would increase processing, transportation, and handling times for fruits and vegetables thus contributing to a negative ecological balance compared to locally grown foods.

Dosimetry"Dose" (short for radiation absorbed dose) is the physical quantity governing the radiation processing of food, relating to the beneficial effects to be achieved. It is measured in the SI unit known as the gray (Gy). One gray of radiation is equal to 1 joule of energy absorbed per kilogram of food material. In radiation processing of foods, the doses are generally measured in kilograys (kGy, 1,000 Gy).

The measurement of radiation dose is referred to as dosimetry and involves exposing dosimeters jointly with the treated food item.Dosimeters are small components attached to the irradiated product made of materials that, when exposed to ionizing radiation,

change specific, measurable physical attributes to a degree that can be correlated to the dose received.

On the basis of the dose of radiation the application is generally divided into three main categories:

Low dose applications (up to 1 kGy)Sprout inhibition in bulbs and tubers 0.03-0.15 kGy. Delay in fruit ripening 0.25-0.75 kGy. Insect disinfestation including quarantine treatment and elimination of food borne parasites 0.07-1.00 kGy

Medium dose applications (1 kGy to 10 kGy) Reduction of spoilage microbes to prolong shelf-life of meat, poultry and seafoods under refrigeration 1.50–3.00 kGy. Reduction of pathogenic microbes in fresh and frozen meat, poultry and seafoods 3.00–7.00 kGy. Reducing the number of microorganisms in spices to improve hygienic quality 10.00 kGy

High dose applications (above 10 kGy)These doses are above those currently permitted in the USA for commercial food items by the FDA and other regulators around the world.The European Union countries allow dried herbs and spices to be irradiated to a maximum dose of 15 kGy which is equal to an "overall average dose" of 10 kGy. Sterilization of packaged meat, poultry, and their products that are shelf stable without refrigeration 25.00-70.00 kGy